Metabolic Intelligence: Budget-Native Compute from Tag to Campus

David Charlot, Open Interface Engineering
Product white paper · OpenIE research.openie.dev
https://research.openie.dev/products/mei/ · PDF https://research.openie.dev/pdfs/mei.pdf

Measurement. Product white paper, kept apart from the research catalog. Not a journal final. Energy figures from the software reference are OpCounter analytical estimates, not board power. We have not synthesized or metered an FPGA board for this white paper. No fabricated citations.

> Designed for Klere (klere.ai). Energy is the binding constraint. The metabolic envelope is the physical condition under which the best answer is obtained. It is not a cheaper answer, and not a Faustian trade. Edge and a resource-optimized central datacenter, not an uncapped hyperscale campus. Dedicated water, power, and cooling plants are not required by design (pack evidence: microwatt-class tags; grid-following lean campus; not invented liters). Research owns Metabolic Intelligence. Klere owns the product. Shannon, Landauer, and Value of Information are established results. Estimates are not board measured_j.

Product white paper. It is kept apart from the research catalog. The research papers are Thermodynamic Bounds and an Optimal Tempo for Goal-Holding Agents, Metered Commit Gates and Task Satiation vs. Aggregate Rebound in AI Demand.

Abstract

Metabolic Intelligence is a budget-native class of computer intelligence. The envelope is the physical condition under which the best answer is obtained. It is not a Faustian choice, and it is not the cheapest answer. Answers that were thought impossible become reachable by metabolic information science: budget-native obtain, binding classifiers, physics-informed features, and test-time close. Grid levers and routing do not degrade the certified answer. They refuse or defer work that would violate the envelope. The answer that commits is the best one. The class is not a transformer update. It sits with Logical Intelligence (energy-based settle), World Labs (spatial world models), and other post-transformer programs as a next wave: intelligence whose first coordinate is the joule envelope, not the next-token corridor.

Designed for Klere. Metabolic Intelligence is designed for **Klere** (klere.ai), Open Interface Engineering's product home for the MEI class. This research study owns the law and class definition. Klere owns the product embodiment. As of this writing the public site is a thin access / energy-efficient AI framing; it is not a shipped feature catalog. Tag, campus, and obtain-stack surfaces named below are **Klere roadmap** unless this study labels them as a **evidence pack**. Product thesis in one line: sell **edge** and **resource-optimized central** under an explicit joule

envelope (not an uncapped transformer hyperscale campus), and keep evidence-pack items separate from roadmap packaging. No invented meters, board `measured_j`, or liters of water saved.

Deployment class. MEI is designed for the **edge** and for the **resource-optimized central datacenter** (a lean, grid-following, budget-capped plant), **not** for the uncapped hyperscale campus that presupposes dedicated cooling loops, dedicated generation, and large ecosystem water/power uptake. At the energy budgets this class targets, those hyperscale support plants are **not required**: that is a **design consequence** of budget-native operation, grounded in pack evidence (µW-class tags under a metabolic schedule; grid-following lean campus with ordered levers). It is **not** a claim of measured liters of water saved, nor a board `measured_j` reading. Estimates remain estimates; software reference path: no FPGA board result claimed.

Class properties, stated plainly:

1. **Physics-informed.** Features and actuators respect sensor kinetics, heater physics, Landauer floors, and grid constraints as established results, not optional add-ons.
2. **Test-time inference.** Binding, routing, and lever selection happen under the live budget; the decision is shaped at serve time.
3. **Super learning / sequential learning.** New contexts and classes can be absorbed under the same budget without a full pretrain replay.
4. **Zero-shot capable under grammar.** When Lookup or Formula covers the coordinate, the system answers without opening a generative leaf.
5. **Not pre-training required** as the default path. Abundance updates, family indices, and obtain-routers learn and decide under local cost; large corpus pretrain is residual, not the substrate.

Dual-scale proof in this study (edge + resource-optimized central, not uncapped hyperscale):

- **Edge / tag scale.** Binding classifiers on cold-chain gas tags. Cortex-M4 instruction counts (emulator-measured): enzyme families with early exit take 1,723 instructions vs 2,451 for an int8 MLP at matched accuracy. On SmellNet sequential learning (50 foods, five groups, no revisit), enzymes keep 77.7% vs 18.4% for the MLP. A metabolic heating schedule draws about 10 µW total (datasheet + schedule model): heater about 3.0 µW, MCU sleep about 6.9 µW on STM32U3; about eight years on a CR2032. Classifier decisions are about 92 nJ; the sleep floor dominates on the metabolic schedule. At this envelope, dedicated cooling plants and campus-scale water loops are not in the design. Coin-cell physics is the plant. Chicken-spoilage Anwar results are internal only and are **not published** here (no licence).
- **Resource-optimized central (lean campus).** A metabolic grid service for a **budget-capped** 100 MW AI plant (grid-following, not uncapped hyperscale), tested end to end against a real OpenADR 3.1 VTN and a real MQTT DSX Flex broker. Plant is a model; protocols are real. With node sleep available: zero time over the grid cap across the closed scenario set. Headline lever order: pause deferral → slow per-user speed → route smaller

model → lower effort → battery → shed. The service is how a lean central stays inside shared power and water ecosystems without requiring a private uncapped cooling/power campus.

Metabolic Intelligence (MEI once in this abstract) owns the **budget envelope and actuators** that make companion floors bind. Mixture of Limits chooses the gear. Notational Intelligence owns certify-before-commit. Satiation owns stop when Value of Information (VoI) is zero or budget refuse fires. This study owns sleep/heater schedules, OpenADR/MQTT service levers, and the $J(m|q)$ obtain-router that prices look up → calculate → model → guess under λ at stake.

Hardware dual-map: the class runs today on GPUs and MCUs (von Neumann), and points at post-von Neumann substrates (neuromorphic, analog / in-memory, photonic, reversible / adiabatic) where energy-based and latent settle map natively. Estimates remain estimates. Software reference path: no FPGA board result claimed; package `measured_j` only when Metered.

The pair. Current fields map AGI to the algorithm and not to paired hardware and software. Metabolic Intelligence refuses that split. Computers are hardware. Software is applied engineering written for that hardware. AGI-grade reliability is the pair. The hardware lottery (Hooker, 2020) is a documented pattern: ideas look like progress because they fit available chips. Mixture of Limits navigates gears that a fabric can host. The argument is The pair, not the algorithm.

Terms used here

Metabolic Intelligence is the AI product class. Klere is hardware and the product home. The software reference implementation is the software path, without board package `measured_j`. Value of information, evidence classes, Periodic Stack, and the board-result flag are on the shared glossary. Living companions: /living/mei/. Campus headline “zero time over cap” always carries the condition already stated in the scenario table: with node sleep available, under the stated plant model.

The pair, not the algorithm

Current AI fields map AGI to the algorithm. They do not map it to paired hardware and software. The model is treated as the intelligence. The chip is treated as a rack the model happens to occupy. The programming language is treated as a neutral notation. Metabolic Intelligence refuses that split. AGI-grade reliability is the pair. It is not the algorithm alone.

Computers are hardware. Software is applied engineering written for that hardware under constraints. An algorithm with no device is a notation. A device with no procedure is a material. Reliability that holds on a whiteboard and fails on the part that must run it is not the grade. The pair either holds on a named host or it does not.

Human-machine interfacing is the tradeoff that writes the pair. Interfacing a person to a machine, and writing algorithms in human-created programming languages, is a tug of war among physics, applied mathematics, and materials science. Physics prices irreversible erasure and data movement (Landauer, 1961; Horowitz, 2014). Applied mathematics is the procedure a person can write, certify, and stop. Materials science is which device can hold that procedure. A language that hides the device does not cancel the tradeoff. It collects the cost as a runtime, as memory traffic, and as a meter that no longer matches the joule.

The hardware lottery

Sara Hooker (2020) named this fact in *The Hardware Lottery* (arXiv:2009.06489; later *Communications of the ACM* 64(12), 2021, doi:10.1145/3467017). A research idea wins because it is suited to the available software and hardware, not because it is superior to alternative directions. The lottery is which ideas look like progress because they fit the chips that already exist. Domain-specialized hardware raises the cost of any direction that does not fit those chips. The field then writes the winner down as an algorithmic result. The winner was the fit. That observation is documented. This study does not reopen it.

The instance in front of every datacenter is exact attention written as a GPU kernel. Dao, Fu, Ermon, Rudra, and Ré (2022) made attention IO-aware: tile Q, K, and V so the $N \times N$ matrix is not materialized in high-bandwidth memory, and fuse the steps into one CUDA kernel that uses on-chip SRAM (*FlashAttention*, arXiv:2205.14135). The algorithm changed because the memory hierarchy of the chip demanded it. Their wall-clock figures stay theirs. They are not this pack's `measured_j`. The mechanism is the point. Attention looked like the path of progress because that kernel was written for the GPU already in the rack.

Availability, accessibility, capability

Mixture of Limits is not a software-only order. Navigation is Lookup, then Formula, then Solver, then Model last, because each gear exists only where a fabric can host it. Three tests decide the host. All three are required.

1. **Available.** The device is manufactured, and a public description of what it executes exists. A whiteboard substrate is not available.
2. **Accessible.** A person can reach the device through a programming language and a runtime that exist: CUDA, a Cortex-M kernel library, a Wasm engine, an FPGA toolchain, a vendor SDK. The reach is the interface tax (§12): data movement, the shim, precision, latency, and meter honesty.
3. **Capable.** The device can execute that gear under the precision contract. `never_lower_precision` either holds or the work is refused. A faster device that returns an uncertified answer is not capable of the commit.

A gear that fails one test is not run on that host. The router does not promote the next gear in order to hide a missing host. $J(m|q)$ prices the host that passed the three tests. The estimator of J is itself Lookup or Formula and is capped so meta-compute cannot eat the envelope B.

Lookup is a memory, not a model

Lookup closes when the coordinate is already in a table the host can address. Horowitz (2014) places practical CMOS energy in moving that data, far above the Landauer ideal. The arithmetic is not the cost. The capable hosts are the ones that already hold an addressable array.

Host	What is available and accessible	What it is capable of	What the host does not price
CPU	Ordinary addressable memory. The language is the host ISA.	A registry or table hit. Movement is the joule.	The joule of the table hit
Wasm	W3C WebAssembly Core Specification, version 3.0, TR/wasm-core (consulted 2 Oct 2026). Linear memory is a contiguous byte array. Load and store are in the spec.	A table inside that array, in a browser or a standalone engine.	Energy. The specification has no energy model
MCU	Flash and SRAM on the Cortex-M tag. The metabolic schedule is the duty cycle around the hit.	A known context or a known refuse, on-device, without an off-tag model.	Board joules. An emulator count is an instruction count
FPGA	On-chip LUT and block RAM, when a card is already in the rack.	A fixed table, if the router stays off the card.	Bitstream energy

The GPU is available and accessible for Lookup and is the wrong first host. A kernel launch moves weights through HBM to recover a coordinate a table already held. That launch is interface tax, not navigation.

Formula is the ALU the language actually emits

Formula closes when a closed form measures the same quantity. Lemma A still holds: at equal or lower variance the lower-energy mechanism wins for every non-negative stake. The host is whichever ALU the language emits, not whichever accelerator won the lottery.

Wasm's numeric instructions are the accessible form of that ALU inside a sandbox: 32- and 64-bit integer and IEEE-754 operations, specified as a virtual instruction set with no architectural joule (same W3C core specification). On a CPU the same closed form is scalar or SIMD code. On a Cortex-M the integer ALU is the capable unit. The bit-enzyme variant in this pack uses no multiplies. The formula is integer arithmetic on that ALU. A GPU can evaluate the same expression. It becomes the host only when the closed form is the residual leaf and the launch still fits B. Otherwise the formula stays on the CPU or the Wasm engine, and Model is not opened.

Solver is settle on a device that can hold the contract

Solver is iteration or physical settle, not a smaller transformer. The capable host depends on the mechanism.

On a CPU, a short certified iteration is available and accessible in ordinary code. That is the host Appendix B already assigns the obtain-router. No new meter is attached to it.

On analog in-memory and Hopfield-class substrates, settle is the device. Dense associative memory, memristor Hopfield networks, and probabilistic in-memory energy-based models are already cited in this study (refs 14, 15, 29, 30, 37, 38). Chen et al. (*Nature Communications*, 30 Sep 2025, doi:10.1038/s41467-025-63794-4) ran an ALBERT inference on a 14 nm analog chip. That demonstration shows a transformer-shaped workload on an analog device. It does not show Mixture of Limits, and it is not this pack's `measured_j`.

A CUDA attention kernel is not a solver. It is a Model leaf that won the lottery. Calling it a solver because it iterates inside a kernel misnames the gear.

Model last is the kernel the chip already runs

Model is the residual gear. It runs where a parametric kernel is available, accessible, and capable, and only after Lookup, Formula, and Solver have failed to cover the coordinate.

GPU. The accessible stack is CUDA. The algorithm that actually occupies it is IO-aware attention, as FlashAttention specified for HBM versus on-chip SRAM, and the serving kernels around it. Selective state-space models (Gu and Dao, 2023; Dao and Gu, 2024) change the recurrence and still ship as GPU kernels. They do not change the host. Field curves for that host are already labeled in this study: InferenceX and MLPerf Inference v6.1 report throughput and derived power. Those figures stay measured-kernel or looked-up. They are not package `measured_j`. NVIDIA Dynamo 1.5 is a vendor power actuator. It is not obtain routing. The interface tax on this fabric is HBM movement, the serving shim, the kernel's numeric format, and token latency.

MCU. The accessible algorithm on a Cortex-M is an integer kernel, not a datacenter transformer. Lai, Suda, and Chandra (2018) published CMSIS-NN, int8 and int16 network kernels for Arm Cortex-M (arXiv:1801.06601). MLPerf Tiny v1.4 (Jul 2026) is the public meter class for MCU inference energy. Both are other groups' results. Neither is this pack's `measured_j`, and neither is the binding classifier. The enzyme comparison this pack stands on is emulator-measured on a Cortex-M4: 1,723 instructions with early exit versus 2,451 for an int8 MLP at matched accuracy, and the SmellNet sequential protocol. CMSIS-NN quantizes. On a commit path, `never_lower_precision` holds or the work is refused. Quantization is not an optimization of a certified quantity.

FPGA. Umuroglu et al. (FPGA 2017, arXiv:1612.07119) mapped binarized networks onto streaming FPGA datapaths: binary operations, on-chip parameters, per-layer folding. That is the published pairing of a low-precision model leaf to LUT fabric. Their platform watts and frame rates stay theirs. This study did not place or route a design and did not meter a card. No FPGA board result claimed. A bit-enzyme or a lookup table is the leaf a card can hold. The router stays on the CPU. A card that narrows a certified quantity to meet a timing constraint is refused.

Neuromorphic, analog, photonic. These devices are available as already cited. They are accessible through a vendor SDK, which is the runtime tax. They are capable of a residual leaf whose mechanism matches the substrate, and not of the obtain-router. Intel's Hala Point description (17 Apr 2024: 1,152 Loihi 2 processors, stated maximum 2,600 W) is a vendor system figure, not a per-answer joule. Ahmed et al. (*Nature*, 9 Apr 2025, doi:10.1038/s41586-025-08854-x) report ResNet, BERT, and an Atari workload on a photonic processor. Tunheim et al. (arXiv:2501.19347) report 8.6 nJ per MNIST frame on a 65 nm Tsetlin accelerator; that nanojoule is their silicon measurement, not OpenIE `measured_j`, and the Tsetlin machine is not the enzyme. Innatera Pulsar-class parts publish audio and radar milliwatts, not a gas-tag schedule. Shipping neuromorphic gas or odour nanojoule figures are not in this citation set.

What the order is doing

Lookup is first because addressable memory is the available, accessible, capable host on every computer this study names, and Horowitz already priced the movement. Formula is next because the ALU the language emits can evaluate a closed form without a kernel launch. Solver is next where iteration or settle fits a host that holds the contract. Model is last because its capable hosts are the specialized devices, and the lottery made those devices look like the definition of intelligence.

Klere (klere.ai) is the designed home because that order is the runtime, not a translation into someone else's stack. The interface tax is lowest there. It is not zero: data movement still costs joules. Metabolic Intelligence is not confined to Klere. The same gears run on the GPU fleet, the Cortex-M tag, the CPU and Wasm host, the accelerator card, and the substrates just cited. The host changes $\mu(p, H)$ and the tax. It does not change the utility of what commits. A host that can only return a worse answer is not used.

The envelope B is physics. It is the condition under which the best answer is obtained. It is not a cheaper answer. Work that would violate B is refused or deferred. The answer that commits is the best one, in the measured sense of Section 2.6. Deployment of every host above is the edge and the resource-optimized central, not an uncapped hyperscale campus. Dedicated water, power, and hyperscale cooling plants are not required by design for the workloads this class serves within its envelope. Workload equivalence with uncapped operation is not claimed. That is a design consequence of the envelope plus pack evidence. It is not a measured liter count.

Mixture of Limits navigates. Notational Intelligence owns certify-before-commit. Satiation owns the stop when Value of Information is zero or budget refuse fires. This study owns the envelope and the actuators those locks spend. Research owns the class. Klere owns the product. Estimates stay estimates. No FPGA board result claimed. Package `measured_j` is written when a labeled meter returns.

Peer fabric pairing and the best answer (2025 to 2026)

The envelope B is the physical condition under which the **best answer** is obtained. Work that would violate B is refused or deferred. It is not degraded into a cheaper answer, and it is not a Faustian trade of certification for watts.

Field peers now publish the same pairing lesson on other axes. Chen et al. (*Nature Communications*, 30 Sep 2025, doi:10.1038/s41467-025-63794-4) ran ALBERT on a 14 nm analog AI inference chip, transformer-shaped work on analog fabric; their accuracy and device counts stay theirs, not package `measured_j`. Greatorex et al., **TEXEL** (*Nature Communications*, 11 Jul 2025, doi:10.1038/s41467-025-61576-6), ship a mixed-signal neuromorphic processor built so CMOS and beyond-CMOS devices can be paired for on-chip learning. That is availability and accessibility of a fabric before a leaf is claimed capable. Huang et al., **A3D-MoE** (arXiv:2507.19142), pair MoE routing to 3D-integrated HBM and fused attention-MoE datapaths. Chung et al. (arXiv:2601.22076) show inference joules are governed by algorithm-software-hardware latent factors on H100/B200.

What Metabolic Intelligence adds. Those citations are substrate or serving co-design. This class binds one envelope across edge and resource-optimized central, with Mixture of Limits gears hosted only where fabric is available, accessible, and capable (§12 interface tax). Klere (klere.ai) is the designed home. The class is not confined to it. Software reference path: no FPGA board result claimed.

Why this is the future

The pair above is the map this future refuses. Industry and academic messaging still treats transformer scale, plus the hyperscale power and water plant that scale presupposes, as the default future of computer intelligence. That default can be answered. It cannot be answered by a slogan. A skeptic who wants to set Metabolic Intelligence aside has to deny settled physics (Shannon, Landauer, or Value of Information), deny a labeled measurement from this pack, or deny an explicit roadmap this study already separates from evidence pack. This section states the argument. Appendix F answers the dismissals that usually follow. Living diagram: /living/mei/#mei-arg-01.

Energy is the binding constraint. Shannon (1948) gives bits a rate-distortion structure: past a fidelity point, more bits buy little additional fidelity. Howard's Value of Information (1966) prices an observation by the improvement it makes in decision utility, net of cost. When that value is zero on a stated completeness predicate, further spend does not raise the written objective. Landauer (1961) prices irreversible bit erasure at $k_B T \ln 2$ per bit in the ideal model. On a receipt that quantity is a labeled estimate, not a wattmeter. Horowitz (2014) shows that practical CMOS energy is dominated by moving data, far above that ideal. These are settled coordinates. They are not Open Interface Engineering inventions. A future whose first coordinate is uncapped watts has to say which of them fails.

The transformer corridor plateaus as a deployment future. The corridor has produced real products: pretrain a large generative model, then spend tokens and watts at serve time. Quality inside that corridor can keep moving. The plateau is not a claim that models stopped improving, and it is not a fabricated scaling chart. Scaling laws hold. Language model loss falls as a power law in parameters, data and training compute [58]. Compute-optimal training grows parameters and tokens together [59]. Test-time compute extends the corridor at serve time, and a compute-optimal schedule can let a smaller model match a larger one on some problems [60]. The plateau is per task, at $C(z) = 1$, not in model quality. The plateau is this: once a chore is complete, extra tokens do not raise the written objective (Satiation). Past the rate-distortion point, extra bits buy little decision utility (Shannon). The plant underneath still presupposes dedicated cooling, dedicated generation, and large ecosystem water and power. Uncapped import stops being the coordinate that buys the next useful answer. Logical Intelligence, V-JEPA, and World Labs are real post-transformer programs on other axes (constraint commit, latent prediction, spatial world models). They do not make that import optional, and this study does not speak for their results.

The envelope obtains the best answer. Let B be the metabolic envelope: joules on a tag episode, or watts on a lean-campus import cap. B is the physical condition of the answer, not a discount on it. Metabolic information science (budget-native obtain, enzymes, physics-informed features, test-time close) is how answers that were thought impossible become reachable. Mixture of Limits chooses the gear: Lookup, then Formula, then Solver, then Model last. The obtain-router prices the gear as $J(m|q)$ before spend: energy plus a stake-weighted uncertainty term. The estimator is Lookup or Formula and is capped so meta-compute cannot eat B. If that estimate exceeds the cap, the router takes the default gear and does not run a heavy estimator. Binding and lever choice happen at test time inside the live envelope. Levers and routing refuse or defer work that would violate the envelope. They do not degrade the certified answer. The answer that commits is the best one. On the digital-enzyme path, new classes can be absorbed without a full pretrain replay. That sentence is path-scoped (Proposition 7). It is not a claim about V-JEPA or World Labs, which pretrain on large video, and it is not a claim that Logical Intelligence skips pretrain.

Dual deployment removes the hyperscale water and power prerequisite. The class is designed for two plants, not for an uncapped hyperscale campus. At the edge, the plant is a coin cell: a metabolic heating schedule on a cold-chain gas tag, datasheet plus schedule model, about 10 µW class, with MCU sleep dominating heater average power on that schedule (Proposition 3). Instruction counts are emulator-measured (1,723 for enzyme families with early exit versus 2,451 for an int8 MLP at matched accuracy). Sequential learning on SmellNet is a real dataset protocol (enzymes 77.7% versus an MLP at 18.4% across 50 foods, five groups, no revisit). At the resource-optimized central, the plant is a budget-capped, grid-following lean campus: real OpenADR 3.1 and a real MQTT DSX Flex broker, with a plant model for the watts. With node sleep available, the closed scenario set records zero time over the grid cap (Proposition 5). Because those envelopes are the design, dedicated water plants, dedicated power plants, and hyperscale cooling loops are not required. That is a design consequence of the class plus pack evidence. It is not a measured liter count. It is not a board `measured_j`. No workload-equivalence measurement exists yet. The protocol is fixed here. Take one task set with

its completeness predicate and deadlines. Serve it once on the lean campus under its import cap and once on an uncapped reference deployment. Report, for both, the fraction closed inside the deadline, joules per closed task, peak import, and water per closed task from the site's own metering. The claim that hyperscale plants are not required holds where the capped campus matches the reference on closure and deadline at its cap. Until that run, it is a design target.

Klere is the product home. This study owns the law and the class definition. Klere (klere.ai) owns product embodiment. As of this writing the public site is thin access framing, not a shipped feature catalog. Tag, lean-campus, and obtain-stack surfaces are Klere roadmap unless this study labels them evidence pack (Appendix E). Computers are hardware. Software is applied engineering under the envelope. Method is the claim axis. Geography is not.

A reader who grants Shannon, Landauer, and Value of Information, and who grants the pack labels above, still has to name a future in which uncapped hyperscale import is required for the next useful answer. Appendix F is the set of ways that denial is usually phrased, and the evidence class of each reply. Software reference path: no FPGA board result claimed.

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1. Introduction: budget-native intelligence

People should not have to fight over energy to eat or to use AI. That civic sentence is the design brief. Cold-chain tags that watch food should last years on a coin cell. Central plants that serve answers should follow the grid when the grid asks them to cut, without inventing false board joules or collapsing every research law into one paper.

Uncapped hyperscale campuses pull dedicated cooling, dedicated generation, and large ecosystem water and power. Metabolic Intelligence is **not** that deployment class. It is designed for the **edge** and for the **resource-optimized central datacenter**: a lean, budget-capped, grid-following plant. Because the class runs at lower energy budgets by construction, dedicated water plants, dedicated power plants, and hyperscale cooling loops are **not required** for the workloads this class serves within its envelope; workload equivalence with uncapped operation is not claimed. This is stated as a **design consequence** of the budget-native envelope, with pack evidence at both ends (µW tags under a metabolic schedule; OpenADR/DSX lean-campus levers). Do **not** invent liters of water saved or board `measured_j` from that consequence. Software reference path: no FPGA board result claimed.

The dominant construction for computer intelligence still treats the transformer corridor as the substrate: pretrain a large generative model, then spend tokens and watts at serve time. That corridor has produced real products. It is not the only class. **Logical Intelligence** casts reasoning as energy minimization over constraints (Business Wire, 20 Jan 2026). **World Labs** casts spatial intelligence as world models that predict views and dynamics (Atlas, 1 Sep 2026). **Metabolic Intelligence** casts intelligence as **budget-native obtain and actuate**: the envelope is the physical condition under which the best answer is obtained. It is not the cheapest answer. The class is designed for **Klere** (klere.ai): research owns the law; Klere owns the product embodiment. Do not read the thin public landing page as live product features.

Shannon (1948) prices bits under uncertainty as established results. Howard's Value of Information (1966) prices whether another observation is worth its cost for a decision. Landauer (1961) prices irreversible bit erasure in joules at temperature T: $E_{\min} = k_B T \ln 2$ per bit erased in the ideal model. On receipts that Landauer quantity is a **labeled estimate**, not a wattmeter reading. Together they imply floors: past a point, more bits stop buying outcomes that matter for a stated benefit. Metabolic Intelligence embodies those floors as **schedules and actuators** at the edge of a tag and at the meter of a lean central plant.

Formal thesis. Let B be the metabolic envelope in joules (tag episode) or watts / megawatts (lean-campus import). B is the physical condition under which the best answer is obtained. It is not a menu of cheaper substitutes. Metabolic Intelligence returns that answer by budget-native obtain, enzymes, physics-informed features, and test-time close. A stated utility U (accuracy, completeness of a chore, interactive close) is what the committed answer maximizes. Grid-following levers refuse or defer work that would violate B. They do not degrade the certified answer. The answer that commits is the best one. The substrate is binding classifiers and other algorithmic compute under that envelope, plus a grid-following service whose levers keep the envelope. Deployment scopes are **edge** and **resource-optimized central**, not an uncapped hyperscale campus whose first coordinate is unlimited import. Class properties follow: physics-informed features and heaters; test-time binding and routing; sequential / super learning without full pretrain replay; zero-shot close when grammar covers; pretrain as residual leaf, not default. Computers are hardware; software is applied engineering under constraints. AGI-grade reliability is that pair, not the algorithm alone (The pair, not the algorithm).

Three measurement facts constrain every later number. First, curve and ledger watt-hours are **model / measured-kernel / looked-up** composites, labeled by evidence class. Second, tag microwatts mix **datasheet** and **schedule model**; instruction counts are **emulator-measured**. Third, no FPGA board result claimed; package `measured_j` appears only when a labeled meter returns a reading.

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2. Formal problem, definitions, and propositions

2.1 Problem statement

Problem (Budget-native obtain). Given a need q that feeds a decision y, an explicit budget B, and a menu of obtain mechanisms $M = \{\text{Lookup}, \text{Formula}, \text{Solver}, \text{Model}, \text{Guess}\}$ with costs and uncertainty, select mechanism m^* and actuation schedule s^* so the committed answer is the superior one under the envelope B. Work that would violate B is refused or deferred. Stop when further spend does not raise a stated completeness predicate (Satiation) or when policy refuse fires (Notational Intelligence). The envelope is not permission to return a worse answer.

The problem is dual-scale by construction: **edge** and **resource-optimized central**, not uncapped hyperscale:

Scale	Budget unit	Actuators	Success metric in this study
Edge / tag	µW continuous; nJ to mJ per event	Heater duty, MCU sleep, enzyme binding / early exit	Lifetime on CR2032; decisions under schedule; sequential accuracy
Resource-optimized central (lean campus)	MW import cap (grid-following; not uncapped)	Pause, speed, route, effort, battery, shed	Time over cap; interactive served share

2.2 Definitions

Term	Definition in this paper
Metabolic Intelligence (MEI)	Budget-native class. The metabolic envelope is the physical condition under which the best answer is obtained, at the edge and the resource-optimized central (not uncapped hyperscale). Not a cheaper answer.
Hardware-software pair	AGI-grade reliability is the procedure together with the fabric that hosts it. Not the algorithm alone. Each Mixture of Limits gear requires a host that is available, accessible, and capable.
Budget envelope	Explicit joule or power condition B under which the best answer is obtained. First-class physical condition, not a post-hoc telemetry label and not a quality discount.
Binding classifier	Silicon recognition unit with binding site, product, allosteric context switch, Hill cooperativity, abundance as learned weight, and a cost for every binding check. Bit-enzyme variant uses no multiplies.
Metabolic schedule	Duty cycle for sensing and heating that meets a monitoring objective while minimizing average power (heater + sleep + radio).
Obtain mechanism	How an input is acquired: Lookup, Formula / Calculate, Solver, Model, Guess, ordered by Mixture of Limits and priced by $J(m \mid q)$.
Obtain-router	The rule $m^* = \arg\min_m J(m \mid q)$ that selects the mechanism under λ at stake.
Grid-following service	Control plane that accepts OpenADR / DSX Flex signals and applies ordered levers so campus import stays under cap.
Evidence class	Label on a number: real protocol, emulator-measured, datasheet, measured-kernel model, looked-up, calculation, plant model, estimate (M/C/V/R/D/S/P in the pack radar).
Software reference path	No FPGA board result claimed; estimates ≠ package `measured_j`.

2.3 Established results: Shannon, Landauer, VoI

These are not OpenIE inventions. They are settled coordinates this study *uses*.

Law	Statement used here	Role in MEI
Shannon (1948)	Bits under uncertainty have a rate-distortion structure; more bits past a point buy little for a fixed fidelity.	Justifies Lookup / Formula before Model when the coordinate is already known.
Landauer (1961)	Irreversible bit erasure has an ideal floor $k_B T \ln 2$ per bit.	Labeled **estimate** on receipts; not a wattmeter. Motivates treating joules as scarce even when digital inference is free at the money margin.
Howard VoI (1966)	Value of Information is the expected improvement in decision utility from an observation, net of cost.	When VoI ≤ 0 under λ, Satiation / Mixture of Limits name stop; MEI makes stop executable as refuse of heater pulse or campus lever.

Horowitz (2014) documents that practical CMOS energy is dominated by data movement and memory, far above the Landauer ideal. That gap is why Metabolic Intelligence schedules heaters and campus import meters rather than claiming thermodynamic optimality from whiteboard erasure.

2.4 Propositions (claims with evidence class)

Each proposition is a claim. Evidence class is stated. Peer overlap is acknowledged; uniqueness is scoped.

Proposition 1 (Envelope primacy). *An intelligence class whose first coordinate is an explicit budget envelope can bind obtain and actuation at microwatt (edge) and megawatt (resource-optimized central) scales under one thesis, without presupposing uncapped hyperscale cooling/power plants.* Evidence: this study's dual-scale pack (tag schedule + OpenADR lean-campus service). Class: **real protocol** (campus) + **datasheet / schedule model / emulator** (tag). Peer slices: Dynamo power caps (campus only); TinyML (edge only). Unique stack claim: dual scale + OpenADR/DSX + $J(m|q)$ + enzymes + Satiation stop. Ecosystem-resource avoidance (water / dedicated power / hyperscale cooling) is stated as a **design consequence** of that envelope, not as measured liters saved.

Proposition 2 (Enzyme sequential learning). *Under a fixed MCU budget, binding classifiers absorb sequential class groups with far less catastrophic forgetting than an int8 MLP of matched decision cost.* Evidence: SmellNet 50 foods, five groups, no revisit: enzymes 77.7% vs MLP 18.4% (**real dataset protocol**). Peer overlap: AdaTM continual Tsetlin; classical Hopfield / DenseAM pattern add. Do **not** claim universal end of catastrophic forgetting.

Proposition 3 (Sleep floor on metabolic schedule). *On a cold-chain gas tag with metabolic heating (~9.6 readings/day, Oct 2026 parts), MCU sleep current dominates heater average power.* Evidence: BME690 heater ~3.0 µW vs STM32U3 sleep ~6.9 µW (**datasheet + schedule model**). Correction: "heater is always the cost" is false on this schedule.

Proposition 4 ($J(m|q)$ obtain law). *Pricing obtain mechanisms by energy plus λ-weighted uncertainty selects Lookup/Formula when they measure the same quantity and reserves Model for residual leaves.* Evidence: synthesis ledger math + sourced inputs (**calculation + looked-up**);

not board `measured_j`. Peer: PEARL / OmniRouter price *which LLM*; they do not price *which obtain zone*.

Proposition 5 (Campus cap with node sleep). *A 100 MW campus plant model under real OpenADR 3.1 and real DSX Flex MQTT, with node sleep available, records zero time over the grid cap across the closed scenario set.* Evidence: **real protocol + plant model**. Without node sleep, idle GPUs alone can exceed a deep cut (scenario E: 96 min over cap).

Proposition 6 (Free-at-margin ≠ free joules). *Published collapse in money price of digital inference does not identify free energy, free actuation, or unbounded value after a chore is complete.* Evidence: companion Satiation (Epoch series cited there); Landauer / Horowitz as floors. MEI supplies the envelope actuators that make the economic stop physical.

Proposition 7 (Class properties are path-scoped). *No-pretrain, zero-shot, and test-time claims must be scoped by path (enzyme vs latent vs certify EBM). Mixing scopes invents a contradiction.* Evidence: V-JEPA 2-AC zero-shot planning requires large video pretrain (Assran et al., 2025); enzyme online path does not. Logical Intelligence constraint energy ≠ campus joules (Business Wire, 2026).

2.5 Formal lemmas (stated, not board-proved)

Lemma A (Obtain dominance under matched measurement). If two mechanisms m_1, m_2 measure the same quantity q and $E_{m_1} < E_{m_2}$ with $\mathrm{var}_{m_1} \leq \mathrm{var}_{m_2}$, then $J(m_1|q) < J(m_2|q)$ for all λ, $S_q \geq 0$. *Proof sketch:* both additive terms weakly favor m_1. Evidence class: **calculation** (definition of J). Implication: Lookup beats Calculate when the table holds the same coordinate; Calculate beats Model when a closed form exists.

Lemma B (Precision sizing). For fixed λ, S_q and target excess cost ε, the uncertainty that equalizes the λ-term with ε is $\sigma^* = \sqrt{2\varepsilon/(\lambda S_q^2)}$ (ignoring staleness). Spending below σ^* wastes energy; spending above leaves decision risk on the table. Evidence: **calculation**. Implication: satiation of precision is a first-class stop, not a soft habit.

Lemma C (Cache refresh). Under linear staleness $\kappa_q \cdot t$, the refresh interval that balances obtain energy against λ-weighted variance growth is $t^* = 2E/(\lambda S_q^2 \kappa)$ when p_m is small. Evidence: **calculation**. Implication: metabolic schedules on tags and cache TTLs on campuses are the same object at different λ.

Lemma D (Cap feasibility without sleep). If idle import P_{idle} exceeds the grid cap P_{cap}, no combination of speed / route / effort levers that leaves nodes powered can meet the cap. Scenario E is the concrete instance (96 min over cap without node sleep). Evidence: **plant model + real protocol**. Implication: node sleep (or equivalent power-down) is not an optional luxury at deep cuts.

These lemmas are engineering consequences of the definitions. They are not theorems about AGI. Software reference path: no FPGA board result claimed.

2.6 Related work by method (academic corridor)

Method is the claim axis. Geography is not. What this class absorbs is stated as a rule. Everything else in this section is cited only. URLs and dates below were checked on 2 Oct 2026. A vendor joule, a vendor latency, or a vendor accuracy stays on the vendor's page. It is not this pack's `measured_j`.

Absorbed (not citations). Four objects are Metabolic Intelligence.

1. **Physics envelope.** B is joules on a tag episode or watts on a lean-campus import cap. Energy budget is physics: the condition under which the answer is obtained. It is not a discount on the answer, and it is not a Faustian cheaper-answer trade.
2. **Best answer.** Levers and routing refuse or defer work that would violate B. They do not degrade the certified answer. The answer that commits is the best one. Best has a measured form. Fix a task set, a completeness predicate $C(z)$ and a deadline per task. For each system, plot the fraction of tasks closed with $C(z) = 1$ inside the deadline against joules per task. Refusals and deferrals count as misses on that curve. They are not hidden. The campus scenario table gives the measured cost of the envelope in those terms: interactive share served falls from 100% to 52.6% at today's speeds under a 90/95% cut, to 13% under a 90% cut for 10 h, and to 1% without node sleep. A system is best at envelope B when no other system closes more tasks inside the deadline at the same joules. That frontier is the test. A claim of superiority that does not place a point on it is not yet a result. Budget-native obtain, binding classifiers, physics-informed features, and test-time close are how answers that were thought impossible become reachable. That reach is inside the envelope.
3. **Binding classifiers.** Binding site, product, allosteric context, abundance as learned weight, a cost on every binding check, early exit. The unit is this study (§5): emulator-measured Cortex-M4 counts and the SmellNet sequential protocol. Tsetlin machines and hyperdimensional computing, cited below, are edge cousins. They are not the enzyme.
4. **Dual hardware map.** The class runs today on GPUs and MCUs. It points at neuromorphic, analog in-memory, photonic, and adiabatic substrates where energy-based and latent settle can map (Appendix B). The map is operational. A substrate paper is not a second definition of the class. Deployment is the edge and the resource-optimized central. Hyperscale water, power, and cooling plants are not required by design. The interface tax of hosting the class on each existing fabric is §12. Klere is the designed home. The class is not confined to it.

Cited only.

Energy-based models, Hopfield networks, and Dense Associative Memory. Hopfield (1982) casts associative memory as descent on an energy. Krotov and Hopfield (arXiv:1606.01164, 2016) raise capacity with higher-order energies (Dense Associative Memory). Ramsauer et al. (arXiv:2008.02217, 2020) recover softmax attention as a modern Hopfield update. Hoover, Krotov, and coauthors (arXiv:2302.07253, NeurIPS 2023) define the Energy Transformer: attention layers built to minimize a token energy. Gladstone et al. (arXiv:2507.02092, 2025) train

Energy-Based Transformers that assign an energy to an input-prediction pair and predict by gradient minimization of that energy. Bacvanski, Hopfield, Krotov, and coauthors (arXiv:2512.15002, 16 Dec 2025) propose analog RC and crossbar circuits for Dense Associative Memory and analyze constant-time inference from amplifier specifications; that bound is their analysis, not a meter here. Montanari, Bullo, Krotov, and Motter (arXiv:2604.05042, 2026) survey energy-based dynamical models from Hopfield networks through Dense Associative Memory to oscillator machines. Logical Intelligence is the product neighbor on this axis, not these theory papers. Bodnia and Hanin (21 Jan 2026) describe Kona as non-autoregressive at the trace, globally scored, and continuous-state: an energy over partial and complete traces, with Aleph coordinating calls to Kona and language models. Kona 1.0 pilots are the Business Wire account of 20 Jan 2026. Their scalar is constraint energy. It is not campus import joules and not a coin-cell schedule. This study does not adopt their benchmark figures.

Latent prediction and spatial world models. LeCun (version 0.9.2, 27 Jun 2022) names the joint-embedding predictive architecture as a path to autonomous machine intelligence. V-JEPA 2 and V-JEPA 2-AC (Assran et al., arXiv:2506.09985, 11 Jun 2025) predict in a latent video space and, after large video pretraining plus less than 62 hours of unlabeled robot video, plan a robot arm from image goals without a task reward. World Labs Marble (12 Nov 2025) is a generative multimodal world model. Atlas (1 Sep 2026) is their later omni model over text, images, video, and 3D. A latent model is an optional Model-last leaf under Mixture of Limits. Their pretrain stays on their path (Proposition 7). This study does not speak for their results.

State-space efficiency corridor. Mamba (Gu and Dao, arXiv:2312.00752, 2023) is a selective state-space sequence model: linear scaling in length, constant-size recurrent state, no attention block. Mamba-2 (Dao and Gu, arXiv:2405.21060, 2024) states a structured state-space duality with attention. The corridor prices sequence compute inside the transformer-replacement race. It does not price an obtain zone, a heater duty cycle, or a grid cap.

Neuromorphic, analog, and photonic substrates. Intel announced Hala Point on 17 Apr 2024: 1,152 Loihi 2 processors, 1.15 billion neurons, stated maximum 2,600 W. That watt figure is Intel’s. Chen et al. (Nature Communications, 30 Sep 2025, doi:10.1038/s41467-025-63794-4) demonstrated an ALBERT transformer on a 14 nm analog in-memory inference chip. Ahmed et al. (Nature, 9 Apr 2025, doi:10.1038/s41586-025-08854-x) report a photonic processor executing ResNet, BERT, and an Atari reinforcement-learning workload. These demonstrations sit on the dual hardware map. They are not the class. Shipping neuromorphic gas or odour nanojoule figures were not found. Innatera Pulsar remains a vendor milliwatt-class audio and radar part, above the tag schedule near 10 µW (datasheet plus schedule model).

Tsetlin machines and hyperdimensional computing. Granmo (arXiv:1804.01508, 2018) learns propositional clauses with teams of Tsetlin automata. Kanerva (*Cognitive Computation*, 2009) defines hyperdimensional computing on high-dimensional random vectors. Tunheim et al. (arXiv:2501.19347, 31 Jan 2025) report an all-digital 65 nm Tsetlin accelerator at 8.6 nJ per MNIST frame. Hyperdimensional Tsetlin machines (arXiv:2406.02648, 2024) combine the two encodings. Cite them as sparse edge cousins: online learning and nJ-class decisions on the tasks they publish. They do not supply the OpenADR obtain-router or the dual-scale thesis.

Open post-training recipes. Trillium Labs (trilliumlabs.org), Nathan Lambert and Tom Zick, was unveiled on 2 Oct 2026. Lambert's note the same day says the nonprofit is building open post-training recipes and intends to expand into open infrastructure for recursive self-improvement, reward hacking, and multi-agent systems. WIRED the same day (Will Knight) reports the initial focus as post-training: publishing experiment details so others can study and replicate them. Those sources describe a program. They do not contain a recipe, a dataset, a configuration, or a model release. None is inferred. Trillium Labs is the open corridor of the frontier transformer stack. It is not Metabolic Intelligence and not a Klere product.

Envelope actuators, not a class. PEARL, OmniRouter, and Energy-Aware LRM (2025 to Jan 2026) price which language model answers. NVIDIA Dynamo 1.5 (18 Sep 2026), InferenceX, MLPerf, Emerald DSX Flex, and the AI Energy Management Alliance are serving and grid actuators (§11.4). Metabolic Intelligence binds those actuator shapes to Mixture of Limits obtain zones. The actuators are not the intelligence class.

Companion law stays companion law, named in full: Mixture of Limits navigates, Notational Intelligence commits, Satiation stops. This study is the envelope those floors spend.

2.7 Evidence grammar (how to read every table)

Tag in this paper	Meaning	When it can support a sell claim
real protocol	Talked to a real VTN / broker / schema	Protocol interoperability yes; plant watts no
emulator-measured	Cortex-M4 instruction counts from ELF under emulator	Relative enzyme vs MLP cost yes; board joules no
datasheet	Vendor electrical tables	Schedule model inputs
measured-kernel model	AI Configurator kernels + published power ratios	Wh/answer bands as estimates
looked-up	Artificial Analysis medians, public blogs	Baseline speeds
calculation	Central differences, J algebra, CoreMark→nJ	Algebraic consequences
plant model	Campus thermal/power simulator	Scenario comparisons under stated assumptions
estimate	Ledger fuse, capacity life	Always banded; never `measured_j`

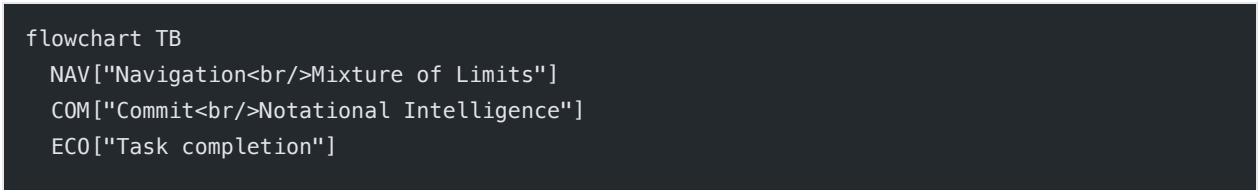
No FPGA board result claimed is sticky. Package `measured_j` appears only when a labeled meter returns a reading.

3. Companion papers and OpenIE map

Four studies on this catalog compose. They do not collapse into one paper. A fifth study, Spell Check is Global, is the automation path. The envelope obtains that answer. It does not discount it. Klere is the product home of this class, not a prison.

Shared thesis, stated once for every paper. Energy to run is the only true metric of computer intelligence. All other factors collapse to zero. The gate form defines what that means. Among runs that pass the task's completeness predicate $C(z)$ within the stated budget and deadline, systems are ranked by joules to completion, J^*. Correctness, deadlines and task size are the gate, not terms in the score. Parameters, tokens, benchmark scores and prices carry zero weight in the ranking (Agency, Section 1). A price trend is a separate claim, labeled as one. Epoch AI reports that the cost of a given level of AI performance has fallen about 47 percent per quarter since 2023, about 13 times per year (Emberson and Roodman, *The plunging price of thought*, 22 September 2026). If that trend continues, energy becomes the dominant marginal cost of inference. That is a cited trend with a condition, not a measured collapse. Spell Check is Global is an existence proof that some computer intelligence commoditizes to bundled zero price. Estimates are never `measured_j`.

Role	Study	Owns
Navigation	Mixture of Limits	Which gear closes: Lookup → Formula → Solver → Model. Floors: VoI, grammar, energy estimate, certificate, settle-refuse.
Commit	Metered Commit Gates, Part I (Notational Intelligence)	propose → certify → commit|refuse → receipt.
Task completion	Task Satiation vs. Aggregate Rebound in AI Demand	Stop when VoI is zero on completeness C(z), or when budget / policy refuse fires.
Energy budget / metabolic embodiment	This paper (Metabolic Intelligence)	Budget envelope and actuators: tag sleep/heater; campus OpenADR/MQTT levers; $J(m \mid q)$ obtain-router.
Automation path	Spell Check is Global	An existence proof that some computer intelligence commoditizes to bundled zero price: spell check's marginal cost fell to about zero, and it complemented hardware and editors people already have. Access framing, not a census: the many (7B+), not the few who rent frontier datacenters (<500M).



```
  MET["Energy budget<br/>Metabolic Intelligence"]
  NAV --> COM --> ECO
  MET -.->|binds floors at tag and campus| NAV
  MET -.-> COM
  MET -.-> ECO
```

3.1 Relation to Mixture of Limits

Mixture of Limits owns *which gear closes*. Metabolic Intelligence supplies the **μ on hardware H** those gears spend. The cascade Lookup → Formula → Solver → Model is the impedance story; enzymes are a budget-native recognition leaf near Formula / cheap Solver; campus Flash routing and speed dials are empirical satiation of tok/s and model size. Software reference path: no FPGA board result claimed; estimates ≠ `measured_j`.

3.2 Relation to Notational Intelligence

Notational Intelligence owns the irreversible commit shape. Metabolic Intelligence does not certify proofs. When a campus lever sheds load or a tag refuses a heater pulse, the *permission* to act remains a Notational / WCA concern; this study owns the energy predicate those commits spend. Logical Intelligence Lean-certified commit is the strongest *peer* certify mechanism (Aleph / PutnamBench materials); it is constraint energy, not grid joules.

3.3 Relation to Satiation

Satiation owns task completion: stop when completeness C(z) holds and VoI is zero, or when budget refuse fires. Metabolic Intelligence makes that stop **executable**:

- Tag: skip the next heater pulse when context surprise is below threshold and lifetime budget is binding.
- Campus: refuse ascending to a larger model or higher speed when J is minimized for the stake.
- Civic: free-at-margin digital inference must not be misread as a licence to burn grid megawatts past done.

3.4 OpenIE map

Surface	URL	Relation
Research	research.openie.dev	This study is `/products/mei/`. Companions: Mixture of Limits, Notational Intelligence, Satiation, Spell Check is Global.
Stack	stack.openie.dev	Family map; MEI is embodiment, not another stack card.
Compute	compute.openie.dev	Primitive table the cascade navigates; metabolic actuators spend primitives under μ on hardware H.

Surface	URL	Relation
Synthesis	synthesis.openie.dev	Cost surface $E(x) = \sum \theta(p) \cdot \mu(p, H)$; this study extends obtain cost as $J(m \mid q)$.
Verify	proof.openie.dev	Cousin energy law for verified artifacts.
Product	klere.ai	**Klere**: product home of the MEI class. This study owns the law/class; Klere owns embodiment. Public site is thin access framing (not a feature list). Tags / campus / stack named in §10 are Klere surfaces or roadmap.

Living figures: /living/mei/ (Manim clips + pack charts + teaching tables). Klere positioning: /about/mei/. Software reference path keeps package energy Estimated unless Metered.

---

4. Standing obtain rule and the J(m|q) synthesis router

Standing rule (David Charlot): **Don't calculate what can be looked up. Don't guess what can be calculated. Don't spend hard-guessing effort on an easy guess.**

The OpenIE synthesis surface prices computing an answer:

$$E(x) = \sum_{p \in P(c(x))} \theta(p) \cdot \mu(p, H)$$

Metabolic Intelligence raises resolution on *how each input is obtained*. For need q feeding decision y, mechanism m has cost:

$$J(m|q) = E_m + \lambda \cdot \tfrac{1}{2} \cdot S_q^2 \cdot \mathrm{var}_m(t)$$

$$\mathrm{var}_m(t) = (\sigma_m^2 + \kappa_q \cdot t) \cdot (1 + 3 \cdot p_m), \quad m^* = \arg\min_m J(m|q)$$

Symbol	Meaning	Evidence
E_m	Energy of the obtain mechanism	Stack formula on the mechanism's primitives
S_q	Elasticity $d \ln y / d \ln q$	**Calculated** by central differences; never assumed
λ	Joules at stake in the decision	Decision scale (one request; campus-year)
σ_m, κ_q, t, p_m	Uncertainty, staleness rate, age, zone-mis-judge chance	Looked-up spread, derivation, or labeled guess

What falls out of the function, not as slogan:

- Lookups dominate when they measure the same quantity.

- Cheap crude options win when λS_q^2 is small.
- Required precision: $\sigma^* = \sqrt{2\varepsilon/(\lambda S_q^2)}$.
- Refresh a cache at $t^* = 2E/(\lambda S_q^2 \kappa)$.
- The estimator of $J(m|q)$ is Lookup or Formula. Its cost is capped so meta-compute cannot eat B. If the estimate exceeds the cap, take the default gear. Do not run a heavy estimator.

Ledger outputs (synthesis pack; math + sourced ledger; **not** board `measured_j`):

Output	Point	p10 to p90	Measure next
Wh per V4 Pro answer at today's speed (GB300)	0.20 Wh	0.044 to 0.68	Meter one GPU pool for a day
Work-per-MW gain, speed matching human chat	1.92×	1.19 to 2.93	Pro speed; human share of tokens
Same gain on Vera Rubin	2.0×	1.2 to 3.4	Metered Rubin chat curve

Fused public proxies for output length (400 / 630 / 1,115 tokens) give 655 at log-σ 0.585, no tighter than a 500-token guess. The router keeps the guess for campus planning and takes the lookup per request. Only the gateway's own counters settle it. Evidence class: **model + looked-up + calculation**.

```
flowchart LR
  L[Look up] --> C[Calculate]
  C --> M[Model]
  M --> G[Guess]
  J["argmin J(m|q)"] --> L
  J --> C
  J --> M
  J --> G
```

---

5. Binding classifiers

A **binding classifier** (earlier drafts said "digital enzyme"; tables keep the short label "enzymes") is a silicon recognition unit with a binding site, a product, an allosteric context switch, Hill cooperativity, an abundance that acts as the learned weight, and a cost for every binding check. A bit-enzyme variant uses no multiplies. The unit is algorithmic compute under a local energy cost, not a smaller transformer.

Result	Number	Evidence class	Where
Instructions / decision, Cortex-M4, 12 features	Enzymes + early exit 1,723; int8 MLP 2,451; first enzyme pool 6,948	Emulator-measured	`mcu/`

Result	Number	Evidence class	Where
Accuracy, same test (3 seeds)	Enzymes 82.9 / 86.9 / 88.7%; MLP 82.2 / 78.0 / 87.4%	Real protocol on simulated features	`benchmarks/`
Late-life accuracy, physics-invariant features (days 90 to 180)	LDA 94.9%, MLP 93.4%, enzymes 90.0% (from 80.8%), Tsetlin 89.1%	Simulation + classifier	`benchmarks/`
SmellNet, 50 foods, 5 sequential groups, no revisit	Enzymes 77.7%; MLP without replay or regularization 18.4% (forgetting floor, not a continual-learning baseline)	Real dataset protocol	`benchmarks/`
SmellNet cross-day	MLP 85.5%; enzymes 79.8% (same on Cortex-M4)	Real dataset + MCU	`benchmarks/`

Class properties, grounded:

- **Physics-informed.** Drift-invariant / physics-informed features added about 9 to 11 points of late-life accuracy for enzymes, bit enzymes, and Tsetlin machines. Sensor kinetics and humidity/heater physics are first-class inputs, not post-hoc regularizers.
- **Sequential / super learning.** Enzymes keep usable accuracy while learning foods one group at a time without revisiting. The MLP collapses under the same protocol (77.7% vs 18.4%). That is the super-learning property under a budget: absorb new context without a full pretrain replay. The MLP here has no replay and no regularization, so 18.4% is the catastrophic-forgetting floor. It is not a strong baseline. The fair comparison runs continual-learning methods under the same memory and energy budget: elastic weight consolidation [61], experience replay with a small buffer [62], and iCaRL with its nearest-mean-of-exemplars classifier [63]. Each is scored for retained accuracy and for energy per inference under MLPerf Tiny measurement rules [64]. That comparison is specified and not yet run.
- **Test-time inference.** Binding checks, early exit through the family index, and allosteric context switches are serve-time decisions under abundance and cost.
- **Not pre-training required** as default. Abundance updates and family indices learn locally. Large generative pretrain is not the substrate of the tag classifier.
- **Zero-shot under grammar.** When a Lookup or Formula leaf covers the context (known context state, known refuse), the cascade closes without opening a generative Model leaf. Mixture of Limits states that order; enzymes supply a budget-native leaf when Model is residual. Open entropy goes to the last cascade stage. That order is the architecture.
- **Capacity.** Family index, early exit, and abundance decay are the capacity control. The sequential learning bound is that control. Garbage collection is abundance decay.

Chicken-spoilage evaluation (Anwar & Anwar 2025) has **no licence**. Results from that dataset are **internal only, not published**. Software reference path: no FPGA board result claimed.

---

6. Cold-chain tag budget

On a gas tag the energy goes into **heating and sleep**, not into classifying. One gas reading costs about 0.5 to 27 mJ depending on the part (datasheet). A classifier decision costs about 92 nJ on an STM32U3 (CoreMark 15.5 µA/MHz at 3.3 V; **calculated** from datasheet, not board package `measured_j`).

Metabolic schedule (about 9.6 readings/day), October 2026 parts:

Term	Power	Evidence
Heater (BME690 metabolic)	~3.0 µW	Datasheet + schedule model
MCU sleep (STM32U3)	~6.9 µW	Datasheet
Tag total	~10 µW	Sum of schedule model
CR2032 life, no self-discharge	~8.1 years	Capacity model: 235 mAh to 2.0 V at 21°C, 3.0 V nominal [65]
CR2032 life, ~1% per year self-discharge	~7.5 years	Capacity model [65]
Same tag on nRF54L15 sleep	~5.1 µW	Datasheet swap

Derating. The cell holds about 235 mAh × 3.0 V, about 2.54 kJ, rated at 21°C [65]. At 9.9 µW the tag uses about 312 J a year, so 8.1 years spends the whole rated capacity. With about 1% per year self-discharge, the life is about 7.5 years. A cold-chain tag lives below 21°C, where lithium coin cells deliver less. That derating is not in the model and must be measured at the cold-chain temperature. The 3.0 µW heater average implies a very low duty cycle for a metal-oxide gas sensor that draws milliwatts when on. Whether sensing response holds at that duty is a measurement to make, not a datasheet fact.

Correction (honesty): earlier prose said “the heater is the cost.” True for frequent readings. On the metabolic schedule the **MCU sleep floor** is the larger term. Choose the MCU for sleep current. Innatera Pulsar-class neuromorphic parts publish audio/radar milliwatts, not gas-tag microwatt schedules; they are peers on the post-von Neumann map, not drop-in tag winners today.

Drift upload is a scheduled act inside that same sleep and heater discipline, inside the CR2032 envelope. It is not an extra drain beside the envelope. Heater duty is thermal feedback inside the same metabolic control. The published ~10 µW class figure stays datasheet plus schedule model.

Market timing (rules, not hype): FDA proposed FSMA 204 compliance move to 20 Jul 2028; Congress barred earlier enforcement. EU PPWR applies since 12 Aug 2026. These dates are the **market clock** for cold-chain observability buyers, not a product launch slogan.

7. Campus energy per answer

Method (curves pack): energy per answer ≈ output tokens × all-in kW per GPU ÷ output tokens/s per GPU.

- **Throughput:** NVIDIA AI Configurator 0.12.0 (measured GPU kernels; SGLang; NVFP4). Evidence: **measured-kernel model**.
- **Power per GPU from InferenceX** (tokens/s per chip ÷ tokens/s per MW): GB300 2.12 kW, B300 1.90, B200 1.71, Vera Rubin 3.30, MI355X 2.09. Evidence: **calculated from published measurements**.
- **Served speeds (Artificial Analysis medians):** V4 Pro 79 tok/s/user, V4 Flash 125, Qwen3-VL 48.5. Evidence: **looked up**.

Workload (GB300 unless noted)	Energy	Evidence class
Chat, DeepSeek V4 Pro (2k in / 500 out)	0.32 Wh at 50 tok/s/user; 0.92 at 79; 4.57 at 159	Curves method
Synthesis ledger best estimate today	0.20 Wh (p10 to p90: 0.04 to 0.68)	Ledger fuse
Chat, V4 Flash	49 mWh at 50; 180 at 159	Curves
Agentic turn, V4 Flash (32k in, 28k cached / 1k out)	0.16 Wh at 96	Curves
Video clip, 8 s 720p (Wan 2.2)	390 Wh	Curves / model

Plateau levers (field + model, not board `measured_j`):

- Vera Rubin NVL72 moves the knee of the speed-energy curve from about 100 to about 175 tok/s/user (InferenceX / MLPerf; **measured** field reports). Speed matching human chat is worth about 2.0× on Rubin vs about 2.9× on GB300 (ledger).
- Flash-first routing (V4.1 Flash answering most human chat) plus speed matching: about 2.3 to 4.4× more work per MW (efficiency pack; **model + looked-up**).
- Deeper speculative decoding (accept length 2.8 modeled): about 1.55× cut in today's base-mix energy per output token; speed matching then adds only about 1.4 to 1.8×.

These are **test-time** levers under a live budget: which model answers, how fast it streams, how much draft speculation to accept. They are Metabolic Intelligence at campus scale, not a new pretrain run.

Bands matter. Point estimates without p10 to p90 invite false precision. Software reference path: no FPGA board result claimed.

8. Metabolic grid service (resource-optimized central)

The service makes a **budget-capped** 100 MW AI plant (a **resource-optimized central datacenter**, not an uncapped hyperscale campus) follow grid caps. Lever order on deferrable work: **pause deferrable work → slow per-user speed → route to a smaller model → lower reasoning effort → battery → shed**. Those levers do not degrade the certified answer. Certified and interactive work runs to close or to refuse. Deferrable work is what pauses. A breakthrough task is not aborted by a linear spend line. A contract declares the invariants (`never_lower_precision`, `max_deferral`). Grid follow stays inside the contract. Shed is last and only outside the invariant. The answer that commits is the best one. Route and effort move deferrable work. They are not a license to commit a worse answer. The withdrawn claim that Flash routing costs no quality stays withdrawn. The power-model bias loop is live thermal control on this service (plant model: learned +5 MW on a hot day). Grid-following lean operation is how the class stays inside shared power and water ecosystems without requiring a private dedicated cooling/power campus. That framing is a design consequence of the envelope; it is not a measured water-liter claim. Software reference path: no FPGA board result claimed.

Architecture (`datacenter/service/`):

- **Grid in:** OpenADR 3.1 VEN (OAuth2, polling, DEMAND reports), tested against real openleadr-rs VTN. Evidence: **real protocol**.
- **Backstop:** DSX Flex ISV over MQTT, CloudEvents, JWS ES256. Evidence: **real broker + schemas**.
- **Actuators:** Dynamo planner GPU budgets; metabolic gateway (in-flight caps, routing, effort, `x-consumer` speed tiers); Kueue / SGLang pause; Redfish node sleep; battery PCS.
- **Learning:** per-pool speed curve by recursive least squares; power-model bias loop (learned +5 MW on a hot day in the plant model).
- **Plant:** campus model. Evidence: **model**. Protocols and schemas: **real**.

Scenario (100 MW GB300 campus)	Time over cap	Interactive served	Note
A to C: 60/70% day; notice / feedforward / no notice	0	100%	Notice mostly saves battery
D: 90/95%, fast campus (159 tok/s/user)	0	100%	All knowledge chat routed; effort low
E: 90/95% **without** node sleep	96 min	1%	Idle GPUs alone exceed a 10 MW cap
L to M: today's looked-up speeds, 60/70 and 90/95	0	100% / 52.6%	Speed already spent cannot be spent again
N: no notice, no battery (PJM-style 10 min)	7.5 min	99.8%	Under cap within 225 s
O: 60% cut for 10 h	0	100%	Deadlines kept

Scenario (100 MW GB300 campus)	Time over cap	Interactive served	Note
P: new cap every 5 min for 3 h	0	98.8%	ERCOT-style
Q: recovery ramp 2 MW/min	0	100%	Max rise 2.9 MW/min vs 29 to 47 unlimited
R: 90% cut for 10 h	0	13%	Video deadline vs interactive (policy)

In every run: no actuator call failed; VTN accepted all OpenADR reports; no DSX message rejected. Headline: **zero time over the grid cap whenever node sleep is available**. Correction honesty: "100% served through a 90/95% cut" assumed a fast campus and one curve; with per-pool curves it is about 94%, from today's speeds about 53%. The **cap held** in all of them.

The measured cost of the envelope. Committed answers keep their certificate in every scenario. The envelope's cost is availability and deferral, and the table measures it in the plant model under real protocols. Interactive share served is 100% in scenarios A to D, O and Q. It is 52.6% at today's looked-up speeds under a 90/95% cut (L to M), 13% under a 90% cut for 10 h (R), and 1% without node sleep (E), with 96 min over the cap. Deferred and refused work is counted, not hidden.

AEMA (Emerald AI, Google, NVIDIA; 16 Sep 2026) is a field alliance; **no published AEMA protocol spec** as of the 1 Oct 2026 radar. Dynamo 1.5 power-aware scaling (18 Sep 2026) is a deployment annotation, not obtain-routing. OpenIE wires those actuators *inside* Mixture of Limits intelligence.

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9. Economic argument: joules as scarce resource

9.1 Free at the margin for digital chores

Satiation states the economic fact: published money price of a fixed inference performance has fallen sharply (Epoch AI series cited there). For digital chores whose output is information and whose completeness test is finite, synthesis can be **free at the margin** in money terms. That phrase does **not** mean joules are free, that a plant may move without a certificate, or that care labor has been automated away.

Metabolic Intelligence accepts that money-price collapse and asks the physical question Satiation leaves to the envelope paper: *who spends which joules after the chore is done, and who is refused?*

9.2 Two scarcities, and the ecosystem that uncapped campuses pull

Scarcity	What binds	Who feels it
Money price of tokens	Collapsing for fixed benchmarks	Buyers of API seats; can look "free"
Grid / actuation joules	Import caps, interconnection queues, heater mJ, battery cycles	Ratepayers, farms, cold-chain operators, campus neighbors
Ecosystem support (water, dedicated power, hyperscale cooling)	Uncapped campus design that pre-supposes private cooling loops and generation	Shared basins, ratepayers, neighbors of hyperscale builds

People should not have to fight over energy to eat or to use AI. The civic framing is not marketing copy. It is the reason tags that watch food must last years on a coin cell, and lean centrals that serve answers must shed watts when the grid asks, without lying about meters.

Design consequence (not a liters-saved claim). Metabolic Intelligence targets energy budgets so low that dedicated water plants, dedicated power plants, and hyperscale cooling loops are **not required** for the workloads this class serves within its envelope; workload equivalence with uncapped operation is not claimed. Evidence in this pack is the envelope itself: ~10 µW tag schedules (datasheet + schedule model) and a grid-following lean campus under OpenADR/DSX (real protocols; plant model). Do **not** invent gallons or liters of water saved, nor upgrade the claim to board `measured_j`. The statement is class design + pack evidence, not a field water meter. Software reference path: no FPGA board result claimed.

9.3 Work-per-MW and Wh/answer as estimates with bands

Quantity	Point	Band	Evidence	Do not claim
Wh / V4 Pro answer (today, GB300)	0.20 Wh	0.04 to 0.68	Ledger fuse	Board `measured_j`
Work-per-MW vs today (speed match)	1.92×	1.19 to 2.93	Efficiency + looked-up speeds	Universal 4 to 12×
Same on Vera Rubin	2.0×	1.2 to 3.4	Ledger transfer	"30×" without speed condition
Flash-first + speed match	2.3 to 4.4×	model band	Efficiency P4	"Routing costs no quality"

These are **estimates with bands**. Software reference path: no FPGA board result claimed. Metering one GPU pool for a day is the highest-VoI next measurement in the synthesis ledger.

9.4 Satiation stop as economic actuator

When completeness C(z) = 1 and VoI is zero, further Model spend is a cost with no increment on the written objective. Metabolic Intelligence implements that stop as:

1. **Obtain refuse:** do not open Model when Lookup/Formula closes J.
2. **Speed refuse:** do not stream above human-read rate when the consumer is human (x–`consumer` tiers).
3. **Grid refuse:** pause / shed when import cap binds, ordered by least service lost per MW.
4. **Tag refuse:** skip heater when surprise is below threshold and lifetime budget binds.

Stop is success. That is Satiation's Economic Reality; this study shows the actuators.

9.5 Landauer as labeled estimate, not a cheat

Landauer's $k_B T \ln 2$ appears on analytical receipts as a **labeled estimate**. It is not a claim that OpenIE boards operate at the ideal floor. Horowitz (2014) places practical CMOS far above that floor. The economic argument needs only the weaker statement: joules remain scarce after tokens become cheap; schedules and meters are how that scarcity is governed.

9.6 Rebound, care, and the civic bound

Direct rebound (Gillingham, Rapson, and Wagner, as treated in the Satiation companion) is the extra use of a service when its effective price falls. Free-at-margin tokens invite rebound in engagement metrics. Metabolic Intelligence does not estimate a structural rebound parameter. It installs **hard stops**: completeness refuse, budget refuse, and grid cap. Those stops are design rules, not fitted elasticities.

Care and food are not software goods. Baumol-style cost disease (cited in Satiation) warns that sectors with weak productivity growth absorb expenditure even when token prices fall. Cold-chain tags exist so food loss and food energy are measured under a coin-cell envelope, not so a generative model can narrate spoilage without a sensor. Lean centrals that shed watts on OpenADR signals exist so ratepayers are not forced into a false choice between "AI" and "keep the lights on," and so shared water and power ecosystems are not presupposed as private hyperscale plants. The civic bound is: **eat and compute under shared joules, with meters and schedules, without invented board claims or invented liters saved.**

9.7 Cost surface from synthesis to metabolic spend

OpenIE synthesis prices an answer as $E(x) = \sum \theta(p) \cdot \mu(p, H)$. Metabolic Intelligence adds:

Layer	Object	Scarcity coordinate
Synthesis	Primitive mix on hardware H	Analytical / looked-up μ
Obtain router	$J(m \mid q)$ over Lookup→Guess	λ at stake + variance
Tag schedule	Heater + sleep + radio	μW continuous; CR2032 life
Lean campus service	Ordered levers under P_{cap}	MW import (capped); interactive served

The same standing rule threads all four layers. Software reference path keeps every layer's joules labeled until Metered.

9.8 Resource-optimized deployment vs uncapped hyperscale

Deployment	Energy budget posture	Ecosystem support presupposed	MEI stance
Edge (tag)	µW continuous; coin-cell life	None beyond the cell and board	**In scope**: evidence pack for schedule / enzyme evidence
Resource-optimized central	MW import **cap**; grid-following levers	Shared grid; no private uncapped cooling/power campus required by class design	**In scope**: evidence pack for OpenADR/DSX + scenarios
Uncapped hyperscale campus	Unlimited import until interconnection / cooling bind	Dedicated cooling, dedicated generation, large water uptake	**Out of class**: not the MEI design target

The market value of the class is not "we save *N* liters." It is: buyers who need answers under an explicit envelope get a deployment that **does not require** the hyperscale support plant. That is the plain voice for the economic argument: design consequence + pack evidence; estimates ≠ `measured_j`.

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10. Market value proposition

10.1 Who buys what

Vehicle. Buyers purchase through **Klere** (klere.ai), the product home designed around Metabolic Intelligence. This paper sells the class and the evidence packs; Klere is the commercial vehicle. The design is **edge** and **resource-optimized central**, not an uncapped hyperscale campus build.

Buyer	What they buy (via Klere)	Why now (market clock)	Evidence class for timing
Cold-chain / food ops (edge)	Microwatt gas tags + enzyme recognition under metabolic schedule	FSMA 204 enforcement barred before 20 Jul 2028 (FDA proposed); EU PPWR since 12 Aug 2026	**R** (rules)
Resource-optimized central / colo / lean campus energy ops	Grid-following lean service: OpenADR 3.1 VEN + DSX Flex MQTT + ordered levers, not uncapped hyperscale cooling/power plants	Emerald DSX Flex commercial path (1 Jun 2026); AEMA alliance (16 Sep 2026, **no spec yet**)	**C/V**

Buyer	What they buy (via Klere)	Why now (market clock)	Evidence class for timing
Software / platform buyers	Enzymes + $J(m \mid q)$ router + metabolic gateway (speed tiers, route, effort)	Flash-first routing and Dynamo power annotations make test-time levers native	**M/V**

10.2 Product slices (honest scope)

Label each slice as **evidence pack** (evidence in this study) or **Klere roadmap** (product packaging not claimed shipped). Klere is the vehicle; this paper is not a SKU sheet.

Slice	What it is	Evidence pack (this study)	Klere roadmap (not claimed shipped)	Ecosystem posture
Edge / cold-chain tags	Enzyme recognition + metabolic heater schedule on coin-cell MCUs	MCU instruction table; SmellNet sequential protocol; tag budget script (~10 µW class, datasheet + schedule model)	Commercial tag SKU / field install / device ops	µW envelope → dedicated cooling/water plants **not in design**
Resource-optimized central	Grid-following lean campus under import cap	OpenADR 3.1 VTN + DSX Flex MQTT demos; scenario table A to R; ordered levers	Commercial flexibility / operator product	Shared grid; **no** private uncapped cooling/power campus required by class design
Obtain / stack surface	$J(m \mid q)$ router + metabolic gateway under Mixture of Limits	Router math; synthesis ledger bands; software reference companions; standing obtain rule	Packaged gateway / DX beyond `mol.yaml` / `mol run`	Software spend under explicit envelope; the last cascade stage

1. **Edge pack.** Binding classifier / bit-enzyme classifiers; family index; physics-informed features; tag budget model for BME690 / ZMOD4410 / SGP41 + STM32U3 / nRF54L15. Not a neuromorphic gas-odour nJ claim (shipping figures **not found**).
2. **Resource-optimized central / lean campus service.** OpenADR 3.1 + DSX Flex-shaped control; gateway with `x-consumer` speed tiers; plant-model tested scenarios. Plant is a model; protocols are real. Sold as grid-following lean central, **not** as an uncapped hyperscale campus that requires dedicated water/power plants.
3. **Obtain stack.** $J(m|q)$ router as Mixture of Limits embodiment; satiation stop wired to refuse codes.

GTM slices (who first). Sequence is engineering, not hype: (A) publish the class with software reference honesty → (B) show protocol demos to lean-campus energy buyers → (C) meter one GPU pool before any Wh/answer upgrade → (D) licence edge pack into Klere packaging when ops buyers are ready → (E) wire Dynamo / DPS when vendor hooks exist.

10.3 Competitive differentiation

Vs transformer hyperscale (primary market foil). The default build is still: pretrain a frontier transformer, then serve tokens under interconnection and cooling that assume an uncapped campus. That path pulls dedicated generation, dedicated cooling loops, and large ecosystem water/power uptake. Klere / MEI refuses that as the design target. The product thesis is **edge + resource-optimized central**: the envelope is the condition of the best answer; the last cascade stage; levers that refuse or defer rather than degrade the certified answer; coin-cell tags. Competitive claim that is allowed: *hyperscale support plants are not required by design at these envelopes*. Competitive claim that is forbidden without meters: *we save N liters / N MW of cooling water*, or *Wh/answer as board* `measured_j`.

Peer	Their axis	Metabolic Intelligence / Klere difference
Transformer-only / uncapped hyperscale stacks	Pretrain → serve tokens under unlimited import + dedicated cooling/power	Envelope-first; the last cascade stage; edge + resource-optimized central; hyperscale support plants not required by design
Logical Intelligence	Constraint energy + formal commit	Joule envelope + obtain routing + dual scale; LI energy ≠ grid joules
World Labs / V-JEPA	Spatial / latent world models	Budget-native enzymes + campus grid service; latent optional; no-pretrain scoped to enzyme path
Pure neuromorphic vendors (e.g. Innatera Pulsar)	Efficient non-transformer silicon at edge	Cousins at edge; lack campus $J(m \mid q)$ + OpenADR thesis; Pulsar mW audio/radar ≠ tag µW schedule
Dynamo / DSX / AEMA	Power and grid actuators	Actuators **inside** Mixture of Limits intelligence; AEMA has no published spec yet
Tsetlin / HDC	Efficient propositional / hypervector silicon	Edge cousins; measured Tsetlin ASIC 8.6 nJ/MNIST frame (Tunheim et al., Jan 2025); lack campus thesis

Honest roadmap vs evidence pack. If a buyer asks “what works today?”, answer with the pack rows (emulator instructions; OpenADR/MQTT demos; schedule model; ledger bands). If they ask “what does Klere ship?”, answer roadmap and do not upgrade evidence class. Software reference path: no FPGA board result claimed.

10.4 Buyer journeys (three packs)

Vehicle for all three journeys: **Klere** (klere.ai). Research study = class + evidence; Klere = product embodiment. Evidence pack vs roadmap is labeled per journey.

Journey A: Cold-chain operator (Klere edge / tags surface). Problem: FSMA 204 / PPWR timing raises the cost of blind legs in the chain; coin-cell tags must last years. Offer via Klere: metabolic schedule (~10 µW class on Oct 2026 parts, datasheet + model) + enzyme recognition

with sequential learning for new SKUs without full MLP retrain. **Evidence pack:** MCU instruction table; SmellNet sequential protocol; tag budget script. **Klere roadmap:** commercial tag SKU / field install packaging. Not offered: Anwar chicken accuracy; neuromorphic gas nJ; board `measured_j`.

Journey B: Resource-optimized central / lean campus energy lead (Klere campus surface). Problem: interconnection and peak stress; AEMA / DSX language is rising; operators need levers that preserve interactive work **without** building an uncapped hyperscale cooling/power campus. Offer via Klere: OpenADR 3.1 VEN + DSX Flex MQTT backstop + ordered levers with scenario pack (zero time over cap with node sleep). Ecosystem-resource avoidance (water / dedicated power / hyperscale cooling) is a design consequence of the envelope, not a liters-saved pitch. **Evidence pack:** real VTN / broker tests; scenario table A to R; recovery-ramp redesign note. **Klere roadmap:** commercial flexibility product and operator SKU. Not offered: AEMA latency/ramp numbers; plant watts as meter readings; invented water liters; "100% served through 90% cut" as universal.

Journey C: Platform / stack buyer (Klere stack surface). Problem: serving stacks route models but do not price obtain zones; transformers stay default. Offer via Klere: $J(m|q)$ router + metabolic gateway (`x-consumer` speed tiers, route, effort) under the Mixture of Limits product path, embodied as Klere stack surfaces. **Evidence pack:** synthesis ledger bands; efficiency pack work-per-MW ranges; standing obtain rule; software reference companions. **Klere roadmap:** packaged gateway / DX productization beyond `mol.yaml` / `mol run`. Not offered: Wh/answer as `measured_j`; "routing costs no quality."

10.5 Pricing and packaging honesty

Claim in a pitch	Allowed form	Forbidden form
Tag lifetime	"~8 years on CR2032 under stated schedule (capacity model)"	"8.00 years measured on shelf"
Enzyme vs MLP	"1,723 vs 2,451 instructions (emulator)"	"30% less board joules"
Campus flexibility	"0 time over cap with node sleep in closed scenarios (plant model + real protocols)"	"Guaranteed 0 over any grid event"
Work per MW	"1.92× point, 1.19 to 2.93 band (ledger)"	"2× guaranteed on your fleet"
Differentiation vs LI	"joule envelope + dual scale vs constraint energy"	"we beat Kona on proofs"
Ecosystem resources	"dedicated water/power/hyperscale cooling not required by design for the workloads served within the envelope (µW tags; lean grid-following central)"	"saves X liters / MW of cooling water" without a meter

10.6 Go-to-market sequencing (engineering, not hype)

Sequencing is for **Klere** as product home; this paper remains the class publication.

1. **Publish the class** (this paper + SOTA brief) with software reference honesty; point product inquiries to klere.ai without overclaiming the thin public site.
2. **Ship protocol demos** (OpenADR/DSX test harness) as interoperability evidence (evidence pack → Klere campus roadmap).
3. **Meter one pool** (highest VoI ledger item) before any Wh/answer sales claim upgrades evidence class.
4. **Licence edge pack** into Klere packaging and seek chicken-dataset permission only if publish is required. Until then, keep Anwar numbers internal.
5. **Wire Dynamo 1.5 / DPS** as live actuators when vendor hooks are available; until then keep them noted, not claimed live.

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11. SOTA at the metabolic frontier

11.1 Class definition

Metabolic Intelligence is the budget-native class: the metabolic envelope is the physical condition under which the best answer is obtained. It does not trade the certified answer down to save joules, and it does not take uncapped transformer FLOPs or uncapped hyperscale import as the coordinate. Architecturally it sits with Logical Intelligence-style moves *past* basic transformers (energy-based scoring and constraint commit, latent world models, non-autoregressive or non-attention primitives), but binds the envelope at **two scales at once**: **edge** (microwatt tags) ⇔ **resource-optimized central** (budget-capped, grid-following lean campus), not the uncapped hyperscale campus that presupposes dedicated water, power, and cooling plants. The unit of computation is the **binding classifier** (recognition, allosteric context gating, abundance as weight, local learning and decay, costed binding check) and related algorithmic compute, not ever-larger nets. Ecosystem-resource avoidance is a **design consequence** of those budgets + pack evidence; it is not a measured liters-saved claim.

Class properties (stated as operating claims; peer overlap cited honestly below):

Property	MEI grounding in this pack	Honest peer overlap
Super / sequential learning	SmellNet enzymes 77.7% vs MLP 18.4% across five groups without revisit (**real protocol**)	AdaTM continual Tsetlin; classical Hopfield / DenseAM pattern add. Do not claim universal end of catastrophic forgetting.
Test-time inference	Binding, early exit, $J(m \mid q)$ route, campus lever dial under live budget	DenseAM energy descent; Hopfield recall; JEPA MPC planning; Logical Intelligence constraint solve. Test-time ≠ free or pretrain-free.
Zero-shot under grammar	Lookup → Formula → Solver first (Mixture of Limits); refuse when covered	**V-JEPA 2-AC** zero-shot robot planning without task reward is a strong peer. OpenADR demos here are **protocol-tested**, not "zero-shot AGI."
No large pre-train	Enzyme / bit-enzyme **online** path on MCU	Tsetlin / Hebbian AM cousins. **Do not** attribute this to JEPA, World Labs, or Logical Intelligence fron-

Property	MEI grounding in this pack	Honest peer overlap
required		tier stacks; they pretrain heavily. Scope the claim to the enzyme path.
Physics-informed	E-nose kinetics, drift-invariant features (+9 to 11 late-life points); heater / sleep schedule; Landauer as labeled estimate	PINN lineage; analog EBM as physics substrate. Physics features ≠ thermodynamic optimality without meters.

Class, not a peer list. Section 2.6 absorbs four objects into Metabolic Intelligence: the physics envelope, the best answer, binding classifiers, and the dual hardware map. The table below cites the rest. Klere (klere.ai) embodies the absorbed class on the edge and the resource-optimized central. Neighbors keep their axis. None of them is this class, and none of this is a geography. The combination here is edge and resource-optimized central, an OpenADR 3.1 and DSX Flex-shaped lean service, the $J(m|q)$ obtain-router, binding classifiers, and Satiation stop, with hyperscale water, power, and cooling plants not required by design. Software reference path: no FPGA board result claimed.

11.2 Method taxonomy (must-cite; method-first, not geography)

The prose, URLs, and dates are in §2.6. This table is the index. Binding classifiers, the physics envelope, the best answer, and the dual hardware map are not rows: they are the class. Checked 2 Oct 2026. Do not promote a cell into `measured_j`.

Method (cited only)	Canonical cite	Cite for	Do not absorb
Energy-based models / Hopfield / Dense Associative Memory	Hopfield 1982; Krotov and Hopfield, arXiv:1606.01164 (2016); Ramsauer et al., arXiv:2008.02217 (2020); Energy Transformer, arXiv:2302.07253 (NeurIPS 2023); Energy-Based Transformers, arXiv:2507.02092 (2025); analog Dense Associative Memory, arXiv:2512.15002 (16 Dec 2025); energy-based dynamical models, arXiv:2604.05042 (2026)	Test-time energy descent. Attention recovered as an energy update.	Their energy is not campus joules and not the tag schedule. Amplifier time bounds are theirs.
Logical Intelligence	Business Wire, 20 Jan 2026; Bodnia and Hanin, 21 Jan 2026, logicalintelligence.com/blog/energy-based-models-for-reasoning	Constraint energy. Non-autoregressive trace. Lean-checkable commit via Aleph.	Not obtain routing. Not dual deployment. Not "no pretrain." Do not copy their benchmark figures.
V-JEPA / World Labs	LeCun, v0.9.2, 27 Jun 2022; Assran et al., arXiv:2506.09985, 11 Jun 2025; Marble, 12 Nov 2025; Atlas, 1 Sep 2026	Latent prediction and spatial world models	Large video pretrain stays on their path.

Method (cited only)	Canonical cite	Cite for	Do not absorb
			Optional Model-last leaf only.
State-space corridor (Mamba, Mamba-2)	arXiv:2312.00752 (2023); arXiv:2405.21060 (2024)	Linear-time selective state. Sequence efficiency versus quadratic attention.	Not an obtain zone. Not a grid cap. Not the class.
Neuromorphic / analog / photonic	Hala Point, Intel, 17 Apr 2024; Chen et al., Nature Communications, 30 Sep 2025, doi:10.1038/s41467-025-63794-4; Ahmed et al., Nature, 9 Apr 2025, doi:10.1038/s41586-025-08854-x	Substrates on the dual hardware map (Appendix B)	Not the software class. Vendor watts are not `measured_j`.
Tsetlin machines / hyperdimensional computing	Granmo, arXiv:1804.01508 (2018); Kanerva, 2009; Tunheim et al., arXiv:2501.19347, 31 Jan 2025 (8.6 nJ per MNIST frame, 65 nm)	Sparse logic and associative edge cousins	Not the enzyme. Not OpenADR.
Trillium Labs	trilliumlabs.org; Lambert note and WIRED, both 2 Oct 2026	Announced program: open post-training recipes on the frontier stack	No recipe, dataset, or release is in those sources. Not Klere.

Power-aware serving and grid flexibility stay in §11.4. They are actuators.

11.3 Positioning vs named peers

Peer	Their axis	Metabolic Intelligence difference
Logical Intelligence	Constraint energy + formal commit	Joule envelope + obtain routing + dual scale
World Labs / V-JEPA	Spatial / latent world models	Budget-native enzymes + campus grid service; latent optional
Dense Associative Memory / analog energy-based models	Physics dynamics for inference	Cited substrate path. Enzymes and the envelope stay this class
Dynamo / DSX / AEMA	Power and grid actuators	Actuators **inside** Mixture of Limits intelligence
Tsetlin machines / hyperdimensional computing	Sparse non-transformer silicon	Edge cousins. The class adds the obtain-router and the OpenADR envelope

Peer	Their axis	Metabolic Intelligence difference
Trillium Labs	Announced open post-training recipes (2 Oct 2026). No recipe in the launch sources	Open corridor of the frontier stack. Not this class

```
flowchart TB
  subgraph classwave [Post-transformer class wave]
    LI[Logical Intelligence<br/>constraint EBM]
    WL[World Labs / JEPA<br/>latent worlds]
    MEI[Metabolic Intelligence<br/>budget envelope]
  end
  subgraph hw [Hardware dual-map]
    VN[GPU / MCU today]
    PVN[Neuromorphic / analog CIM<br/>photonic / adiabatic]
  end
  MEI --> VN
  MEI --> PVN
  LI -.-> PVN
  WL --> VN
```

11.4 Soft peers only: serving and grid actuators

Dynamo 1.5 power-aware scaling, NVIDIA DPS, InferenceX Vera Rubin tok/s/MW curves, Emerald DSX Flex, and the AI Energy Management Alliance are **envelope actuators and field measurements**. They do not define the intelligence class. Academic joule-aware routers (PEARL, OmniRouter, Energy-Aware LRM) price which LLM answers; they do not price which obtain zone (Lookup → Formula → Solver → Model) and do not bind tag↔campus.

Term hunt (2 Oct 2026): **Metabolic Intelligence** as a named peer class to Logical Intelligence / World Labs was **not found** outside OpenIE / Charlot Lab energy-first framing. That is a positioning fact, not a monopoly claim on every energy paper.

11.5 What not to claim without meters

- Wh/answer, J/token, µW lifetime, or × work/MW untied to pack evidence **M/C** or a named external meter.
- Invented **liters / gallons of water saved** (state ecosystem-resource avoidance as design consequence of budget-native class + pack evidence only).
- “Routing costs no quality” (Flash leads AA composite; trails ~1.5 GPQA).
- “Heater is always the tag cost” (sleep floor dominates on metabolic schedule).
- Rubin “up to 30×” without speed condition (gain is speed-dependent).
- AEMA / DSX Flex latency or ramp numbers (unspecified).
- Shipping neuromorphic gas/odour nJ figures (not found).
- That Logical Intelligence “energy-based” equals grid joules (different energy).

- That MEI needs no pretrain **and** matches V-JEPA zero-shot robotics (incompatible scopes; split by path).
- Chicken-spoilage accuracy in public prose (**no licence**).

Software reference path: no FPGA board result claimed; estimates ≠ `measured_j`.

The gear-to-host pairing is The pair, not the algorithm. The state of the art on each existing fabric, and the interface tax of hosting the class there, is State of the art on every existing compute fabric.

---

12. State of the art on every existing compute fabric

Which algorithm runs on which fabric, under the tests of availability, accessibility, and capability, is stated in The pair, not the algorithm. Metabolic Intelligence is the state of the art of budget-native obtain on every compute fabric that already exists. The state of the art is the class. It is not a silicon leaderboard, and it is not a cheaper answer. Four objects travel with the class onto every host: the physics envelope B, the best answer, binding classifiers, and the dual hardware map. The best answer is the one that commits. Work that would violate B is refused or deferred. It is not degraded. Mixture of Limits chooses the gear: Lookup, then Formula, then Solver, then Model last. The obtain-router prices that gear as $J(m|q)$ before spend. The estimator is Lookup or Formula and is capped so meta-compute cannot eat B.

SLA-bounded levers are the contract already stated as a rule. The invariants are `never_lower_precision` and `max_deferral`. On the resource-optimized central the order is pause, then speed, then route, then effort, then battery, then shed. On the tag the lever is the metabolic schedule. Shed and a worse commit sit outside the invariant. Energy on every host is $E_{\text{mem}} + E_{\text{compute}}$. Instruction counts, kernel time, and device ops are the compute term. Memory movement is in the budget. Horowitz (2014) places practical CMOS energy in that movement. The orders in §13.7 are literature physics. They are not this pack's `measured_j`.

The fabric sets $\mu(p, H)$. It does not set the utility of what commits. Klere (klere.ai) is the designed home and the product home. It is where Metabolic Intelligence works best, because the home is built so the four objects are the runtime. Metabolic Intelligence is not confined to Klere. The same objects run on alternate fabric. A host that can only return a worse answer is not used.

The **interface tax** is what a host charges for not being that home. Five terms, on every row below:

1. **Data movement.** Crossings the designed home does not make: HBM, a sandbox copy, a host-to-card link, on and off a substrate.
2. **API or runtime shim.** CUDA, a vendor HAL, a WASM runtime, a toolchain, a neuromorphic SDK. The shim stays. The law sits in front of it.
3. **Precision.** The host's numeric format. `never_lower_precision` either holds or the work is refused. Rounding a certified quantity is not an optimization.

4. **Latency.** The host's clock: tokens per second, an instruction path, a sandbox call, a pipeline fill, a settle time. It is not a licence to commit early with a worse answer.
5. **Meter honesty.** What the host will report as joules. Curves, datasheets, vendor watts, and other groups' silicon figures keep their evidence class. Package `measured_j` is written when a meter on this pack returns. No FPGA board result claimed.

Deployment on every row is the edge and the resource-optimized central. An existing GPU fleet is a budget-capped host. It is not an uncapped hyperscale campus. Dedicated water, power, and hyperscale cooling plants are not required by design for the workloads this class serves within its envelope. Workload equivalence with uncapped operation is not claimed. That is a design consequence of the envelope. It is not a measured liter count.

Fabric	What runs there	Interface tax	Versus Klere-native	Path to the state of the art on that fabric
Klere native	Enzymes, $J(m \mid q)$, contract levers, tier meters	Lowest. No class-to-runtime shim. Memory energy remains. Meters stay in class	The reference home	The envelope obtains the best answer on the edge and the resource-optimized central
GPU / CUDA stacks	Model-last leaf and campus actuators. The router stays in front of the kernel	HBM movement, serving shim, host precision, token latency. Curve watts are not `measured_j`	The installed fleet and kernel curves. The enzyme schedule is not native. Idle power needs node sleep	Price J before launch. Keep the contract. Count E_{mem}
MCU / Cortex-M	Enzyme leaf. The metabolic schedule is the lever	A bare image pays the HAL, SRAM, and emulator-not-joules. The Klere schedule collapses that tax to physics	Coin-cell proof. A tag alone is not the campus service	Family index, early exit, sleep floor, Lookup before any off-tag model
CPU / WASM	Lookup and Formula. The router host	Sandbox or process copies, runtime shim. No pack joule on a WASM leaf	Grammar closes without an accelerator. No tag plant and no grid plant on the leaf itself	The standing obtain rule. Model last is another fabric
FPGA / accelerator card	Bit-enzyme and lookup datapath, as a mechanism	Host link, toolchain shim, fixed-point under the contract. Meter empty	A deterministic leaf when the card is already there. No right to quote card watts	Router and contract stay on the home. Refuse the link when it would break B
Neuromorphic, analog in-memo-	Residual sparse or energy leaf	Vendor SDK, on-and-off movement, their	A lower $\mu(p, H)$ on a matched leaf. The	The home keeps B and J. The

Fabric	What runs there	Interface tax	Versus Klere-native	Path to the state of the art on that fabric
ry, photonic	only	precision, their latency. Their figures stay theirs	envelope is not inside the chip	substrate does not define the class

12.1 Klere native

Klere is the designed home. The interface tax is the lowest of the six, and it is not zero. Data movement still costs joules. The home does not delete E_{mem}. It refuses to hide that term inside a FLOP count or a token price.

Binding classifiers are the recognition leaf: binding site, product, allosteric context, abundance as the learned weight, a cost on every binding check, early exit. The evidence pack under that leaf is the Cortex-M4 emulator count and the SmellNet sequential protocol (§5). Klere packaging of a tag SKU is roadmap (Appendix E). The law and the pack are not waiting on the SKU. The public site is thin access framing. It is not a feature catalog, and it is not a shim. Research owns the class. Klere owns the embodiment.

The obtain-router is native. $J(m|q)$ runs before spend. Lookup and Formula close a covered coordinate. Solver is next. Model is last. The estimator of J is itself Lookup or Formula. If that estimate exceeds the cap, the router takes the default gear and does not run a heavy estimator. SLA-bounded levers are native on both plants: the metabolic schedule on the edge, and the ordered grid levers inside the contract on the resource-optimized central. Notational Intelligence remains the commit gate. Satiation remains the stop. They are companions. They are not translation layers.

Precision is the contract. There is no foreign quantizer on the commit path. Latency is binding, routing, and lever selection under the live budget. It is not an adapter parked in front of someone else's serving API. Meter honesty is part of the low tax. The home reports evidence class. Datasheet plus schedule, emulator counts, real protocol, plant model, calculation, and estimate stay in their rows. Package `measured_j` stays empty until a labeled meter returns. That empty field is the instrument. It is not a weaker class.

This row is the reference. Every later tradeoff is priced against it. What the home does not pretend to have shipped is an operator console, a field SKU, or a live Dynamo wire-up. Those stay roadmap. The state of the art on the home is the envelope obtaining the best answer at both design plants, with hyperscale water and power not required by design.

12.2 GPU and CUDA serving stacks

Existing datacenters already run CUDA serving stacks. That fleet is a host. It is not the class, and it is not a requirement to build an uncapped campus around it.

The mapping is specific. A transformer kernel is a Model-last leaf or a serving actuator. It is not a binding classifier. Enzyme recognition stays on the edge path, or as a host-side leaf in front of the launch. $J(m|q)$ runs in the gateway. If Lookup or Formula measures the same quantity, Lemma A says the kernel is not the minimizer, and the kernel is not launched. SLA-bounded levers on deferrable work are the §8 order, inside `never_lower_precision` and `max_deferral`. Certified and interactive work runs to close or to refuse. Route and effort move deferrable work. They do not commit a worse answer. The withdrawn claim that Flash routing costs no quality stays withdrawn. NVIDIA Dynamo 1.5 power-aware scaling (18 Sep 2026) is a vendor actuator shape. It is not obtain routing.

$E_{\text{mem}} + E_{\text{compute}}$ on this fabric is dominated by weight and cache movement through HBM and the interconnect. Kernel time is only the compute term. The interface tax is that movement, plus the serving shim. CUDA, the engine used for the curves (SGLang in the AI Configurator method), and the GPU planner price tokens and speed. Metabolic Intelligence prices the obtain zone first. The shim remains in the rack. The law sits in front of the launch.

Precision on the fleet is the kernel's format, including the NVFP4 kernels in the measured-kernel method. The contract forbids lowering precision to buy the cap. Latency on the fleet is tokens per second per user. Looked-up medians and measured-kernel throughput are the host's clock. They are not binding-check latency. Meter honesty: the synthesis ledger's 0.20 Wh per V4 Pro answer at today's speed on GB300 (p10 to p90: 0.044 to 0.68) is calculation plus looked-up inputs. InferenceX-derived all-in power (GB300 2.12 kW, B300 1.90, B200 1.71, Vera Rubin 3.30) is calculated from published measurements. Both are estimates or measured-kernel models. Neither is `measured_j`.

Against Klere-native the fleet is a gain and a loss. The gain is the installed plant, the kernel curves, and a service that already speaks GPU budgets. The pack's campus shape is a budget-capped 100 MW plant model under a real OpenADR 3.1 VTN and a real MQTT DSX Flex broker. With node sleep available, that closed scenario set records zero time over the grid cap (Proposition 5, plant model plus real protocol). The loss is the native enzyme, the coin-cell sleep floor, and a runtime whose first object is the envelope. Without node sleep, idle GPUs alone miss a deep cut (scenario E: 96 min over cap). Lemma D is the mechanism: if idle import already exceeds the cap, levers that leave nodes powered cannot meet it. Node sleep is feasibility. The deployment stays resource-optimized central. Dedicated water, power, and hyperscale cooling are not required by design.

The path to the state of the art on this fabric is the order above. Price $J(m|q)$ before launch. Bind the contract to the live cap. Count HBM movement in the envelope. Refuse or defer work that would violate B. The answer that commits is the best one. That is the metabolic state of the art on an existing CUDA datacenter. It is not a new joule record on GB300 or Rubin silicon.

12.3 MCU and Cortex-M class edge

Cold-chain tags are Cortex-M class parts. This is where the evidence pack measures the enzyme, and where a bare image still pays an interface tax if it is not the Klere schedule.

The binding classifier maps directly. On a Cortex-M4, emulator-measured, 12 features: enzyme families with early exit take 1,723 instructions per decision; an int8 MLP takes 2,451 at matched accuracy. On SmellNet, 50 foods, five sequential groups, no revisit, enzymes hold 77.7% where the MLP falls to 18.4%. That protocol is the sequential-learning property under a fixed budget (Proposition 2). It is not a universal end of catastrophic forgetting. Physics-informed features are part of the same leaf. The bit-enzyme variant uses no multiplies.

$J(m|q)$ on the tag has a short menu. Known context and known refuse close on Lookup or Formula. The enzyme is the residual recognition leaf. A generative model is not the substrate. The no-pretrain sentence is this path only (Proposition 7). SLA-bounded control is the metabolic schedule: heater duty and MCU sleep inside the coin-cell envelope, with thermal feedback on the heater. Drift upload is a scheduled act inside that same envelope. This study does not publish a separate radio joule.

$E_{\text{mem}} + E_{\text{compute}}$: the published 1,723 is the compute term. SRAM movement is in the budget whether or not the emulator printed it. A classifier decision at about 92 nJ is calculated from the STM32U3 CoreMark datasheet figure (15.5 µA/MHz at 3.3 V). It is not a board package meter. On the metabolic schedule of about 9.6 readings per day, the sleep floor dominates the heater: heater about 3.0 µW, STM32U3 sleep about 6.9 µW, tag total about 10 µW, CR2032 life about 8.1 years on the capacity model (datasheet plus schedule model, Proposition 3). The life figure is an estimate. The instruction count is emulator-measured. The accuracy protocol is real.

The interface tax depends on who owns the schedule. A bare Cortex-M image pays the vendor HAL, the RTOS if one is linked, the fixed SRAM crossings, and a meter that stops at instructions. A float copy of the same decision is an extra precision tax. It is not the enzyme the emulator counted. Latency is the instruction path, not a network round trip. Meter honesty is mandatory: emulator counts are not joules. When Klere owns the schedule, this fabric sits inside the home, and the tax collapses to the physics that remains: sleep current, heater physics, and the calculated 92 nJ. The MCU is the edge plant of the product. It is not a second product company. That is why the home is still the lowest tax, including here.

Against full Klere, a tag alone loses the OpenADR and DSX Flex service and the gateway router. Those live on the resource-optimized central and on the CPU host. The class is the two plants. Against an int8 MLP on the same core, the enzyme path is the result this pack stands on. The MLP is the control.

The path to the state of the art on this fabric: family index, early exit, abundance decay; physics-informed features; a schedule chosen for sleep current; Lookup and Formula before any off-tag model; refuse the next heater pulse when Value of Information does not pay. Do not promote the emulator to `measured_j`. Chicken-spoilage numbers are not published.

12.4 CPU, WASM, and lookup-formula hosts

Appendix B places the $J(m|q)$ router on a CPU or gateway today. That host is existing fabric. WASM is the portable sandbox form of the same host: a table and a closed form inside a linear memory, in a browser or an edge gateway. No WASM joule is in this pack. None is invented here.

The native leaves are Lookup and Formula. Mixture of Limits puts them first for a reason that is algebraic. Lemma A: when two mechanisms measure the same quantity, the lower energy at equal or lower variance wins for every non-negative stake. A table that holds the coordinate beats a calculation. A closed form beats a model. An enzyme binary can run as integer code on the CPU. That is a faithful port of the algorithm. It is not the tag's sleep physics, and it is not a GPU kernel.

The SLA-bounded lever on this host is the gateway's cap. The estimator of J is Lookup or Formula and is capped. Over the cap, the default gear runs. A heavy estimator does not. Open entropy goes to Model last, and Model last is a call to another fabric, made only when this host does not pretend to close. $E_{\text{mem}} + E_{\text{compute}}$ on a formula is usually the copy, not the arithmetic. A sandbox crossing into WASM linear memory is E_{mem}. A covered coordinate that never opens a model never pays HBM.

The interface tax is the runtime. Process boundary or WASM sandbox, plus the JS boundary when the host is a browser, plus the gateway RPC when the host is a service. Each copy is data movement. Precision is the table's precision. A sandbox float that rounds a certified quantity violates `never_lower_precision` and is refused. Latency is the call overhead and the cache miss. For a covered coordinate that overhead is still the short path, because the alternative is a model round trip. Meter honesty: this row is software and calculation. There is no package `measured_j` for a WASM leaf.

Against Klere-native the gain is reach. The class runs where there is no GPU, no Cortex-M, and no custom silicon. Answers that were thought to require a model become reachable when the grammar covers the coordinate. That is zero-shot under grammar, scoped to the covered leaf, not a claim about robot planning. The loss is the plants. A WASM table is not a coin-cell tag and not an OpenADR campus. Klere remains the home that binds those plants. The CPU or WASM host is how the same law runs on ordinary computers.

The path to the state of the art on this fabric is the standing rule. Don't calculate what can be looked up. Don't guess what can be calculated. Don't spend hard-guessing effort on an easy guess. Price the residual with $J(m|q)$. The state of the art on a CPU or WASM host is the covered answer under B. It is not a smaller model.

12.5 FPGA and existing accelerator cards

No FPGA board result claimed. This study has no FPGA board-synthesis meter and no package `measured_j` from an accelerator card. This section does not add one. Existing cards are still a fabric the class can be hosted on. The path is a mechanism. It is not a bitstream result.

A bit-enzyme uses no multiplies. Lookup is a table. A binding check with early exit is a fixed datapath. Those shapes are what LUT logic and on-card memory hold. That is the mapping. It is not a place-and-route report. $J(m|q)$ stays on the host CPU. The card does not become the router. SLA-bounded levers stay on the control plane. A card that narrows a certified quantity to meet a power rail or a timing constraint violates `never_lower_precision` and is refused.

$E_{\text{mem}} + E_{\text{compute}}$ on a card is, until a meter says otherwise, the host link plus on-card memory movement. PCIe or on-board DMA can exceed the on-fabric operation. Count that crossing in J or do not call the card. The interface tax is that link, the vendor toolchain, and the driver runtime. Fixed-point width is chosen so the contract holds. Pipeline fill is the latency term. Early exit pays only when the card can return before the full pipeline. This study did not measure that return. Meter honesty is the empty field. A programmed part elsewhere in the catalog is not a Metabolic Intelligence joule.

Against Klere-native the gain, when a card is already in the rack, is a deterministic leaf for a binding check or a lookup, without launching a GPU kernel for a covered coordinate. The loss is the designed schedule, the evidence pack (Cortex-M emulator and real protocols, not a card), and any right to quote card watts. Hyperscale support plants are still not required. A card in a lean host does not create a water plant.

The path to the state of the art on this fabric: keep the router and the contract on Klere or on the CPU gateway that obeys the same law; place only a leaf the card executes without lowering precision; refuse the call when link energy would violate B; leave `measured_j` empty until a labeled meter returns. That is the metabolic state of the art on accelerator cards that already exist. It is not a placed-and-routed energy claim.

12.6 Neuromorphic, analog in-memory, and photonic substrates

These substrates exist. Some of them ship. They sit on the dual hardware map. They are not a second definition of Metabolic Intelligence, and they do not eat the class (Appendix F). Computers are hardware. The class runs on whatever hardware spends the joules. Each citation below was already verified in this study. Each keeps its label. None is this pack's `measured_j`.

Substrate	Already verified in this study	Label
Hala Point	Intel, 17 Apr 2024. 1,152 Loihi 2 processors, 1.15 billion neurons, stated maximum 2,600 W	**Vendor.** Not `measured_j`
Analog in-memory ALBERT	Chen et al., *Nature Communications*, 30 Sep 2025, doi:10.1038/s41467-025-63794-4. A 14 nm analog inference chip running ALBERT	**Their demonstration.** Not this pack's meter
Photonic processor	Ahmed et al., *Nature*, 9 Apr 2025, doi:10.1038/s41586-025-08854-x. ResNet, BERT, and an Atari workload	**Their demonstration.** Substrate only. Not `measured_j`
Tsetlin ASIC	Tunheim et al., arXiv:2501.19347, 31 Jan 2025. All-digital 65 nm accelerator, 8.6 nJ per MNIST frame	**Their silicon measurement.** Edge cousin. Not the enzyme.

Substrate	Already verified in this study	Label
		Not OpenIE `measured_j`

Sparse clause teams, energy descent, analog in-memory multiply, and photonic matrix ops are what those devices compute. A binding classifier is a binding check with abundance and early exit. The Tsetlin machine is a cousin on the edge. It is not the enzyme, and it does not supply the obtain-router or the OpenADR envelope. Analog ALBERT and photonic BERT show a transformer workload running on a new device. They show the device. They do not show Mixture of Limits.

$J(m|q)$ stays on the host. Spike cost, analog settle, and a photonic multiply are $\mu(p, H)$ for a residual leaf. They are not the metabolic envelope B, and they are not campus import joules. SLA-bounded levers do not move inside the neuron, the crossbar, or the mesh. If the device cannot hold `never_lower_precision`, the leaf is refused. $E_{\text{mem}} + E_{\text{compute}}$: these substrates are built to cut data movement, which is the Horowitz term, and that cut is their claim. On the host side of the link, movement remains. This study does not add a measured saving.

The interface tax is the shim and the crossing. A Loihi graph, an analog programming model, a photonic compile, or the Tsetlin accelerator's datapath is the runtime tax. Metabolic Intelligence does not become the vendor SDK. Hala Point's stated maximum of 2,600 W is Intel's system figure. It is a vendor description of a machine. It is not a per-answer joule, and it is not evidence that the host link is free. Precision is the device's precision, including analog variation. The contract holds or the work is refused. There is no rounding discount on the committed answer. Latency is event time, settle time, or time of flight. The SLA is the host's. A faster device that returns an uncertified answer is outside the envelope. Meter honesty is the label column. Shipping neuromorphic gas or odour nanojoule figures are not in this citation set. This study does not invent them. Section 6 already places shipping neuromorphic microcontroller audio and radar parts above the tag's metabolic schedule. That comparison is datasheet plus schedule model against a vendor class. It is not a new cite.

Against Klere-native the gain is a lower $\mu(p, H)$ on a leaf whose mechanism matches the substrate. The gain belongs to the citation that measured or announced it. The loss is everything that makes the class: the envelope, the obtain order, the tag schedule, and the grid service. None of those four citations contain that stack.

The path to the state of the art on this fabric: Klere, or a CPU gateway under the same law, keeps B, $J(m|q)$, and the contract. The substrate executes only the residual leaf that matches. Work that would violate the envelope is refused or deferred. The answer that commits is the best one. State of the art here is that composition. It is not a claim that this pack metered Hala Point, the 14 nm analog chip, the photonic processor, or the 65 nm Tsetlin ASIC.

Adiabatic and reversible machines stay on Appendix B as a pointed path. This section does not add a citation for them.

12.7 One answer, six hosts

The comparison is against Klere-native. It is not against a degraded answer. On every fabric the committed answer is the same object: the best answer under B. Alternate fabric changes the interface tax and changes $\mu(p, H)$. It does not change the utility of what commits. Klere is where the tax is lowest and where the product lives. Metabolic Intelligence is how the same class is the state of the art on the GPU fleet, the Cortex-M tag, the CPU and WASM lookup host, the accelerator card, and the substrates already cited. Confinement to one chip family is false. A Faustian cheaper commit is false. Hyperscale water and power are not required by design.

Software reference path: no FPGA board result claimed. Estimates are not `measured_j`.

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13. Review answered by construction

Metabolic Intelligence does not bargain the answer down. Klere is the product that builds the class. The envelope is the physical condition under which the best answer is obtained. The points below are mechanisms of that class, stated as a rule. Pack numbers stay in their evidence class: emulator counts, datasheet schedules, protocol tests, plant models, calculations. No FPGA board result claimed because meters are the measurement instrument of the solution. Package `measured_j` is written when a meter returns. No shunt joules are invented. Chicken-spoilage Anwar numbers are not published.

13.1 A priori difficulty

The obtain router prices cost before spend. $J(m|q)$ is that estimate. The estimator is a Lookup or Formula gear. Its compute is capped so meta-compute cannot eat B. If the estimate of the estimator exceeds the cap, the router takes the default gear and does not run a heavy estimator.

13.2 Enzyme capacity

Family index, early exit, and abundance decay are the capacity control. The sequential learning bound is part of the enzyme. Garbage collection is decay of abundance. Cortex-M4 counts in §5 are emulator-measured instructions for the family-index path. They are the compute term of that control.

13.3 Non-monotonic utility

Levers do not assume a smooth utility of energy. Certified and interactive work runs to close or to refuse. Deferrable work is what pauses. A breakthrough task is not aborted because a linear spend line was crossed. The answer that commits is the best one. The envelope is not a Faustian discount.

13.4 Two ledgers

Compile-time synthesis energy and the serve-time envelope are both counted. Tag microwatts (§6, datasheet plus schedule model) are the serve-time envelope. Offline synthesis sits on the compile ledger $E(x)$. Dual accounting is the law. Compile joules stay on the compile ledger. Serve joules stay on the serve ledger.

13.5 Contract-bounded levers

A contract declares invariants: `never_lower_precision`, `max_deferral`. Grid follow happens inside the contract. Shed is last, and only outside the invariant. The ordered levers in §8 are that law on the lean campus. They refuse or defer work that would violate the envelope. They do not degrade the certified answer.

13.6 Edge sync inside the envelope

Drift upload is a scheduled act inside the metabolic budget, under the same sleep and heater discipline as the rest of the tag. It is inside the CR2032 envelope. The published schedule (about 10 µW, datasheet plus schedule model) is the envelope those acts share. This study does not add a separate radio joule.

13.7 Energy is memory plus compute

The energy metric is $E_{\text{mem}} + E_{\text{compute}}$. Instruction counts are a compute term only. Memory movement is in the budget. Horowitz (2014) places practical CMOS energy in data movement: on the order of 10 pJ for an internal cache access or a functional operation, and 1 to 2 nJ for a DRAM access, in that plenary's technology frame. Those are literature orders for the physics of moving data. They are not this pack's `measured_j`.

13.8 Thermal and process loop

The control loop is live. On the campus service, power-model bias is learned (plant model: +5 MW on a hot day). On the edge, the metabolic schedule includes thermal feedback: heater duty follows the thermal state, inside the same envelope. That loop is metabolic control.

13.9 Open entropy is the last cascade stage

A bounded domain is the class. Work covered by Lookup, Formula, or Solver closes there. Open entropy goes to the last cascade stage. That order is the architecture.

13.10 Primitive distillation

Grammars and formulas are extracted from traces and installed as Lookup, Formula, or enzyme. Human craft of those leaves is not the long-term path. Sutton's Bitter Lesson (2019) says general methods that scale with compute overtake hand-built features. This class answers that lesson by distillation into Lookup, Formula, and enzyme. It does not answer it with uncapped pretrain.

13.11 Tier meters

Energy is reported by tier. When a meter returns, the package field `measured_j` is that reading. Until then the row stays an estimate, a datasheet, an emulator count, a protocol test, or a plant model. Estimates stay estimates.

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14. Corrections log

Condensed from the Metabolic Intelligence handoff corrections log. Withdrawn claims are stated plainly.

Date	Withdrawn or revised	Correction
2 Oct 2026	FSMA 204 “compliance moved to 20 Jul 2028” as settled	FDA **proposed**; Congress barred enforcement before that date
2 Oct 2026	PJM IRAS “full reduction within 10 min”; effective date 12 Oct 2026	Transmission owner must start and complete reduction within 10 min of PJM quantity; effective date unverifiable, removed
2 Oct 2026	STM32U3 at 12 µA/MHz as CoreMark	That figure is a while(1) loop at 48 MHz; CoreMark is 15.5 µA/MHz (51 pJ/cycle; 92 nJ/decision **calculated**)
1 Oct 2026	“V4.1 Flash routing costs no quality”	Leads V4 Pro on AA composite (39 vs 36); trails ~1.5 GPQA points on knowledge
1 Oct 2026	Recovery ramp as fixed rising line	Lagging actuators jumped 13 to 31 MW/min; redesigned plan+one-step with battery on meter (2.9 MW/min)
1 Oct 2026	100% served through 90/95% cut as universal	Fast campus / one curve; per-pool curves ~94%; today’s speeds ~53%; **cap held** in all
1 Oct 2026	Speed-matching gain 2 to 4× / 4.6 to 12×	Baselines were guesses; looked-up speeds → 1.65 to 2.9×
1 Oct 2026	All-in kW 2.25 / 2.0 / 2.15 (GB300/B300/B200)	InferenceX-derived: 2.12 / 1.90 / 1.71
1 Oct 2026	B300 “as good per Wh at low speed”	Withdrawn; GB300 2.9× B300 tokens/MW at 96 tok/s/user (InferenceX)
1 Oct 2026	Chat answer 5 to 62 mWh from InferenceX J/token	InferenceX counts total tokens (mostly cached input); replaced by AI Configurator curves
1 Oct 2026	Flash at 0.33× Pro energy/token	AI Configurator: 0.04 to 0.15×; routing stronger, speed dial weaker
1 Oct 2026	Edge: “heater is the cost”	Frequent readings yes; metabolic schedule → MCU sleep floor dominates

Date	Withdrawn or revised	Correction
Sep 2026	Enzyme MCU cost estimated in Python	Real counts 2.3 to 2.9× MLP; family index → 0.70×
Sep 2026	Older systems as current SOTA; XOR as two-site MWC	Relabelled or removed

Chicken-spoilage Anwar numbers stay **unpublished** (no licence).

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15. Measurement classes

Evidence classes used in this study: **real protocol**, **emulator-measured**, **datasheet**, **measured-kernel model**, **looked-up**, **calculation**, **plant model**, **estimate**. Estimates ≠ `measured_j`. Software reference path: no FPGA board result claimed.

Tier meters are how the solution reports energy (§13.11). Package `measured_j` is written when a meter returns. Until then the row stays in the class below. Stake order on the synthesis ledger, for the next meter readings:

1. One GB300 pool for a day (energy-per-answer band; campus-scale value of the meter).
2. Vera Rubin chat curve from a metered pool or AISimulate once Rubin-supported.
3. Gateway counters: output tokens per answer; human share (`x-consumer`).
4. V4.1 Flash throughput curve (the curve behind routing, in place of the V4 Flash stand-in).
5. Speculative-decoding accept length per workload from engine telemetry.

Tag figures are datasheet plus schedule plus emulator. Campus figures are curves plus plant model plus real protocols. Stage C package joules are not claimed. No invented board package joules. No invented shunt joules. Chicken-spoilage Anwar numbers stay unpublished (no licence).

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16. Conclusion and relation to Mixture of Limits product

How that class is the state of the art on every fabric that already exists, and what interface tax each host charges relative to Klere, is State of the art on every existing compute fabric.

Metabolic Intelligence teaches a budget-native class: physics-informed, test-time, zero-shot under grammar, sequential / super learning without mandatory pretrain, dual-mapped to GPUs/MCUs today and to neuromorphic / analog / photonic / adiabatic substrates next. The envelope is the physical condition under which the best answer is obtained. It is not a cheaper answer. The mechanisms are stated as a rule in Review answered by construction. Designed for the **edge** and the **resource-optimized central datacenter**, not the uncapped hyperscale campus. At those energy budgets, dedicated water, power, and hyperscale cooling plants are **not required** (design consequence of the class + pack evidence: µW tags; grid-following lean

campus; not invented liters saved). Binding classifiers prove the class at microwatts. The OpenADR / DSX Flex service proves the class at a budget-capped 100 MW lean plant. The $J(m|q)$ router makes the standing obtain rule executable. The economic argument binds free-at-margin digital inference to scarce joules and shared ecosystems. The market value proposition names who buys tags, lean-campus flexibility, and the software stack **through Klere**, without selling unmetered numbers or inventing live product features from a thin landing page. The argument a skeptic must answer is Why this is the future. Dismissals and their evidence class are in Appendix F. Software reference path: no FPGA board result claimed; estimates ≠ `measured_j`.

Companions stay distinct. Mixture of Limits navigates Lookup → Formula → Solver → Model and owns the product path (`mol.yaml` / `mol run` / A1 to A14; openIE-dev/mixture-of-limits). Notational Intelligence owns certify-before-commit. Satiation owns task completion. This study owns the envelope and actuators that make those floors bind when food is watched on a coin cell and when a campus must shed watts without lying about meters.

Product surface: Klere under Mixture of Limits

Metabolic Intelligence is **designed for Klere** (klere.ai). Mixture of Limits decides the gear. Metabolic Intelligence supplies the envelope those floors spend. **Klere** is the product embodiment of the class, on edge tags and in resource-optimized central datacenters. The Mixture of Limits cascade decides the gear; metabolic actuators and the obtain-router decide how hard the envelope may be driven at tag sleep current and at lean-campus import meters. When VoI is zero or the budget refuse fires, Satiation and Mixture of Limits already name stop as success. This study shows that stop is executable with OpenADR reports and node sleep, that enzyme recognition can stay inside a coin-cell envelope without a transformer pretrain, and that the class does not require dedicated water/power/hyperscale-cooling plants as a design consequence of its budgets.

Software reference path: no FPGA board result claimed; estimates ≠ `measured_j`. Living figures: /living/mei/ (Manim dual-scale + obtain-router clips; pack charts; teaching tables). Investor/partner one-pager: /about/mei/. PDF: /pdfs/mei.pdf when regenerated.

Where the products sit

This section holds the product placement that the physics paper no longer carries. It moved here from Sections 1, 2.4 and 8.13 of Thermodynamic Bounds and an Optimal Tempo for Goal-Holding Agents, unchanged in substance.

- **Metabolic Intelligence:** the OpenIE product. It runs on any fabric.
- **Klere:** the hardware.
- **Metabolic Intelligence** computes ι on any fabric.
- **Klere** is the hardware that meters J.
- **Klere** is the hardware that meters J. M0 and M1 stand in for it until Klere meters are on the bench.

- **Metabolic Intelligence** runs the arms on any fabric and computes ι from receipts.
- **Mixture of Limits** is the philosophy. Both selectors in the agency runs are instances of it. Arm 2 is its fixed cascade instance.

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Appendix A. Notation quick reference

Symbol	Meaning
B	Budget envelope (J or W / MW)
$\mathcal{A}(B)$	Obtainable answers under B
U(a)	Stated utility of answer a
q, y	Need and decision
m, M	Obtain mechanism and menu
E_m	Energy of mechanism m
S_q	Elasticity $d\ln y / d\ln q$
λ	Joules at stake
σ_m, κ_q, t, p_m	Uncertainty, staleness rate, age, misjudge chance

Symbol	Meaning
$J(m \mid q)$	Obtain cost
C(z)	Completeness predicate (Satiation)
$E(x)$	Synthesis cost surface
$\mu(p, H)$	Primitive energy on hardware H
P_{cap}, P_{idle}	Grid cap and idle import

Appendix B. Dual hardware map (operational)

Class leaf	Runs today (von Neumann)	Post-von Neumann path	Status in this study
Binding classifier	MCU / Cortex-M4 C	Content-addressable / sparse associative	Emulator-measured instructions
$J(m \mid q)$ router	CPU / gateway service	N/A (software)	Math + ledger
Campus levers	GPU planners, Redfish, MQTT	Analog power fabrics (speculative)	Real protocols; plant model
Latent / EBM optional leaf	GPU ViT / Lean	Analog DenseAM / memristor Hopfield	Cited peers; not OpenIE meters
Photonic / adiabatic	Lab claims	Joule-floor substrates	Cited; not MEI software class

Appendix C. Civic one-pager (for non-specialists)

1. AI answers are getting cheaper in money. Electricity, water, and food cold-chains are not free.
2. **Metabolic Intelligence** means: the energy envelope is the physical condition of the best answer. It is not the cheapest answer. Levers refuse or defer work that would break the envelope. The answer that commits is the best one.
3. On a food tag (the **edge**), that means years on a coin cell and smart heating, not a giant model in the package.
4. On a **resource-optimized central** plant, that means following the grid when asked, keeping interactive work alive, and stopping when the chore is done, not building an uncapped hyperscale campus with its own cooling and power plants.
5. At those lower budgets, dedicated water, power, and hyperscale cooling are **not required by design** for the workloads this class serves within its envelope. Workload equivalence with uncapped operation is not claimed. We do not invent liters saved; we show the envelope and the pack evidence.
6. We label every number: measured, datasheet, model, or estimate. We do not invent board joules.

7. People should not have to fight over energy to eat or to use AI.
8. Product home: **Klere** (klere.ai). Research owns the class; Klere owns embodiment.

Appendix D. Glossary of definitions

Spell-outs and working definitions used in this study. First use in the body also carries a short gloss; this appendix is the lookup table.

Term	Definition
Metabolic Intelligence (MEI)	Budget-native class. The metabolic envelope is the physical condition under which the best answer is obtained, at the edge and the resource-optimized central. Not a cheaper answer. Not a Faustian trade.
Klere	Product home of the MEI class (klere.ai). Research owns the law/class; Klere owns embodiment. Public site is thin access framing unless a feature is labeled evidence pack.
Budget envelope B	Explicit joule or power condition under which the best answer is obtained. Not a quality discount.
Binding classifier	Silicon recognition unit with binding site, product, allosteric context switch, Hill cooperativity, abundance as learned weight, and a cost for every binding check. Bit-enzyme variant uses no multiplies.
Metabolic schedule	Duty cycle for sensing and heating that meets a monitoring objective while minimizing average power (heater + sleep + radio).
Obtain mechanism	How an input is acquired: Lookup, Formula / Calculate, Solver, Model, Guess, ordered by Mixture of Limits.
Obtain-router / $J(m \mid q)$	Rule $m^* = \arg\min_m J(m \mid q)$ with $J(m \mid q) = E_m + \lambda \cdot \frac{1}{2} \cdot S_q^2 \cdot \mathrm{var}_m(t)$.
Grid-following service	Control plane that accepts OpenADR / DSX Flex signals and applies ordered levers so campus import stays under cap.
Resource-optimized central	Budget-capped, grid-following lean campus. Not an uncapped hyperscale plant with dedicated water/power/cooling as design defaults.
Edge / tag	Microwatt-class cold-chain or similar sensing node under coin-cell physics.
Evidence class	Label on a number: real protocol, emulator-measured, datasheet, measured-kernel model, looked-up, calculation, plant model, estimate (pack radar M/C/V/R/D/S/P).
Evidence pack	Evidence artifact present in the OpenIE metabolic handoff pack and cited with its class in this study.
Klere roadmap	Product packaging / SKU / operator surface not yet shipped as a live Klere feature; may reuse evidence pack under separate evidence class.
Estimate	Number that is modeled, looked-up, or curve-derived and not a board package meter.

Term	Definition
Software reference path	No FPGA board result claimed; estimates ≠ package `measured_j`.
Value of Information (VoI)	Howard (1966): expected improvement in decision utility from an observation, net of cost.
Satiation	Companion law: stop when VoI is zero or budget refuse fires (task completion).
Mixture of Limits	Companion navigation rule: Lookup → Formula → Solver → Model. Each gear is hosted by a fabric that is available, accessible, and capable. Not a software-only order.
Hardware lottery	Hooker (2020): a research idea wins because it fits available software and hardware, not because it is the superior direction. In this study the lottery is which ideas look like progress because they fit available chips. Documented observation (Hooker, 2020).
Hardware-software pair	AGI-grade reliability is the algorithm together with the device it runs on. Computers are hardware. Software is applied engineering written for that hardware.
Notational Intelligence	Companion commit gate: certify-before-commit.
OpenADR	Open Automated Demand Response protocol (3.1 VTN used in pack demos).
DSX Flex	Emerald / NVIDIA-shaped campus flexibility MQTT service used as peer actuator shape in pack demos.
AEMA	AI Energy Management Alliance (Emerald AI, Google, NVIDIA). No published protocol spec as of 1 Oct 2026 radar.
Interface tax	What a host charges to run Metabolic Intelligence when that host is not the designed home: data movement, API or runtime shim, precision conversion, added latency, and meter honesty. Klere pays the lowest tax. The tax is not permission to commit a worse answer.
board_metered	Repository honesty flag. True only when a labeled meter returned an FPGA board energy reading for the study. False here.
measured_j	Package joule field populated only when a labeled meter returns a reading (Metered).
Dual ledger	Compile-time synthesis energy and serve-time envelope, both counted. Tag µW is serve-time. Offline synthesis is compile-time.
Contract invariant	Declared bound on levers (`never_lower_precision`, `max_deferral`). Grid follow stays inside it. Shed is outside it and last.
Primitive distillation	Extraction of grammars and formulas from traces into Lookup, Formula, or enzyme.

Appendix E. Claim ledger (evidence pack vs roadmap vs estimate)

Sell these as **separate evidence classes**. Do not upgrade a row without changing its class and citation.

E.1 Evidence pack (cite in sales as pack evidence)

Claim	Evidence pointer	Class
Enzyme families + early exit: 1,723 instr/decision vs int8 MLP 2,451 (matched Cortex-M4 setting)	Pack `mcu/` emulator	Emulator-measured
SmellNet sequential: enzymes 77.7% vs MLP 18.4% (50 foods, 5 groups, no revisit)	Pack `benchmarks/`	Real dataset protocol
Metabolic tag schedule ~9.92 µW total (heater ~2.99, sleep ~6.93) on BME690+STM32U3 model	Pack `tag_budget_2026_results.json`	Datasheet + schedule model
OpenADR 3.1 VTN + DSX Flex MQTT service tested end-to-end	Pack `datacenter/service/`	Real protocol
Ordered levers: pause → speed → route → effort → battery → shed	Pack service + study §8	Real protocol + plant model
Closed scenario set with node sleep: zero time over cap (pack headline)	Pack scenario table / charts	Plant model + real protocol
Standing obtain rule / $J(m \mid q)$ math + ledger bands	Pack `synthesis/` + study §4	Calculation + looked-up
Software reference companions (Mixture of Limits / NI / Satiation)	Catalog papers	Research study

E.2 Klere roadmap (not invented shipped features)

Surface	What is verified underneath	What is still roadmap
Edge / tag SKU	Enzyme + schedule pack evidence	Klere packaging, ops licence, field SKU
Lean-campus operator product	OpenADR/DSX demos + scenario table; contract invariants are class law (§13.5)	Operator console and multi-site packaging
Obtain / stack gateway	Router math, software reference `mol` companions, speed-tier synthesis	Packaged gateway DX beyond research surfaces
Dynamo 1.5 / DPS live actuators	Vendor docs as envelope peers	Live wired power-cap actuators in Klere

E.3 Estimate / labeled model (not board measured_j)

Quantity	Status
Wh/answer curve bands	Curves / measured-kernel model → **Estimated** until Metered
CR2032 lifetime years	Capacity model from datasheet ÷ average µW → **Estimate**
Campus plant watts / time-over-cap charts	Plant model (protocols real) → **Model**
Landauer $k_B T \ln 2$ on receipts	Labeled thermodynamic lower-bound **estimate**
Water / power / cooling "not required by design"	Design consequence of envelope + pack evidence for the workloads served within it, **not** measured liters saved; workload equivalence with uncapped operation not claimed
AEMA latency / ramp	**Forbidden**: no published protocol spec
Anwar chicken-spoilage accuracy	**Internal only**: no licence; not published

E.4 Software reference lock

No FPGA board result claimed. Package `measured_j` only when Metered. Investor/partner prose must keep evidence pack, product packaging, and estimate on separate lines. No FPGA board result claimed names the meter instrument. It does not weaken the class.

E.5 Mechanisms of the class (design law)

These rows are the construction in §13. They are not open work. Evidence class on any number stays with the pack row that measured it.

Mechanism	Law
A priori obtain	$J(m \mid q)$ before spend. Estimator is Lookup or Formula, capped. Over cap: default gear.
Enzyme capacity	Family index + early exit + abundance decay. Decay is garbage collection.
Non-monotonic utility	Certified and interactive work closes or refuses. Deferrable work pauses. No abort of a breakthrough by linear spend.
Dual ledger	Compile-time synthesis and serve-time envelope are both counted. Tag µW is serve-time.
Contract levers	Invariants `never_lower_precision`, `max_deferral`. Shed last, only outside the invariant.
Edge sync	Drift upload is inside the metabolic budget and the CR2032 envelope. No separate radio joule published.

Mechanism	Law
$E_{\text{mem}} + E_{\text{compute}}$	Instruction counts are the compute term. Memory movement is in the budget. Horowitz orders are literature physics, not `measured_j`.
Thermal loop	Campus bias learning (plant model) and edge heater feedback are the control.
Model stage, last in the cascade	Open entropy goes to the last cascade stage. That is the architecture.
Primitive distillation	Traces yield Lookup, Formula, and enzyme. Not uncapped pretrain.
Tier meters	`measured_j` when a meter returns. Estimates stay estimates.
Fabric portability	The same envelope, enzymes, obtain-router, and contract levers are the class on every existing fabric (§12). The fabric sets $\mu(p, H)$ and the interface tax. It does not set a cheaper answer. Klere is the home. The class is not confined to Klere.

Appendix F. Objection ledger

These are the dismissals that try to put Metabolic Intelligence back inside "not the future": a nicer serving stack, a pretrain story, a hardware fashion, a peer that already finished the move, an empty market, or a pile of estimates. Each reply is one paragraph. The evidence class is what has to be denied. None of these replies invent liters, board `measured_j`, or chicken-spoilage numbers. None of them speak for Logical Intelligence, V-JEPA, or World Labs beyond the axis this study actually cites.

F.1 "This is just efficient inference."

Efficient inference prices tokens, speed, or which model answers, under a transformer that is still the substrate. Metabolic Intelligence prices the envelope first, then the obtain zone. Mixture of Limits orders Lookup, Formula, Solver, and Model last. The obtain-router $J(m|q)$ is calculation on that menu, not a leaderboard of large language models. Dynamo 1.5 power caps, PEARL, OmniRouter, and Energy-Aware LRM are real envelope actuators or joule-aware routers. Section 11.4 keeps them as soft peers: they do not bind a microwatt tag to a lean campus, and they do not choose an obtain zone. Denying this reply means showing a published class that already does that stack. Term hunt on 2 Oct 2026 did not find one. That is a positioning fact, not a monopoly on every energy paper. Evidence class: **class definition** plus **peer taxonomy** (study §11), not a wattmeter.

F.2 "You still need a pretrain."

The no-pretrain sentence is scoped to the digital-enzyme path (Proposition 7). On that path, abundance and binding learn under a local cost. SmellNet sequential learning (50 foods, five groups, no revisit) is a real dataset protocol: enzymes hold 77.7% where an int8 MLP falls to 18.4%, without a full pretrain replay of earlier groups. Cortex-M4 instruction counts are emulator-measured, not joules. The same sentence is false if it is pointed at V-JEPA 2-AC or World Labs,

which require large video pretrain for their zero-shot planning results, or at Logical Intelligence's frontier stacks. Split the path or the claim is mixed and should be rejected. Evidence class: **real dataset protocol** (enzyme path only) plus an explicit **scope limit**.

F.3 "Neuromorphic hardware will eat this."

Neuromorphic, analog in-memory, photonic, and adiabatic machines are substrates on the hardware map (Appendix B), not a rival definition of the class. Computers are hardware. The class has to run on whatever hardware spends the joules. It runs today on MCUs and GPUs. It points at post-von Neumann parts where energy-based and latent settle map more natively. Innatera Pulsar, a shipping neuromorphic microcontroller, is about 0.4 to 0.6 mW on audio and radar in the pack notes, far above a duty-cycled tag near 10 µW (datasheet plus schedule model). Shipping neuromorphic gas or odour nanojoule figures were not found. A substrate can lower $\mu(p, H)$. It does not by itself choose Lookup before Model, follow an OpenADR signal, or stop when Value of Information is zero. Evidence class: **vendor** (Pulsar class) plus **datasheet / schedule model** (tag). Not a claim that neuromorphic parts are useless. The interface tax of those substrates, and of the fabrics the class already runs on, is §12.

F.4 "Logical Intelligence already did energy-based models."

Logical Intelligence is a class neighbor, not a predecessor that already owns this envelope. Kona 1.0 pilots (Business Wire, 20 Jan 2026) move constraint energy and Lean-checkable commit into product pilots. That energy is the energy of a constraint landscape. It is not campus import joules, and it is not a coin-cell schedule. Notational Intelligence, the companion commit gate on this catalog, is the Open Interface Engineering surface for certify-before-commit. Logical Intelligence is the strongest peer certify mechanism this study cites. Metabolic Intelligence does not claim their proofs, their PutnamBench numbers, or their pilots. The difference on the axis that matters here is the joule envelope, obtain routing, and dual deployment. Evidence class: **vendor** (their pilots) versus **this pack** (tag schedule, OpenADR / DSX protocol, obtain-router math). Denying the difference requires equating constraint energy with grid joules.

F.5 "World Labs or JEPA is already the post-transformer future."

V-JEPA 2 and V-JEPA 2-AC (Assran et al., 11 Jun 2025) are a strong zero-shot planning peer: latent prediction and robot planning without a task reward, after large video pretrain. World Labs Atlas (1 Sep 2026) continues a spatial world-model corridor. Metabolic Intelligence treats a latent model as an optional Model-zone leaf under Mixture of Limits, not as the enzyme substrate. Their zero-shot result is real and should not be shaved down. It also does not price a grid cap, a heater duty cycle, or Lookup before Model. Using their result to retire a budget-native class mixes two methods. Evidence class: **measured / vendor** on their axis (cited, not re-metered here) and **path scope** on ours (Proposition 7).

F.6 “There is no market.”

The market claim in this study is a buyer and a clock, not a total addressable market. Cold-chain operators face a rules clock: FDA’s FSMA 204 move is proposed, and Congress barred enforcement before 20 Jul 2028; EU PPWR has applied since 12 Aug 2026 (evidence class **R**). Lean-campus energy leads already sit in OpenADR and DSX Flex language; the AI Energy Management Alliance has no published protocol spec as of the 1 Oct 2026 radar, so this study does not sell alliance latency. Platform buyers already route models and still do not price obtain zones. Klere is the commercial vehicle. Evidence pack is what can be shown now (emulator counts, SmellNet protocol, schedule model, OpenADR and MQTT demos, router math). Klere roadmap is packaging, not a shipped catalog invented from a thin landing page. Evidence class: **rules** for timing, **evidence pack** for the offer, **roadmap** kept separate. No invented revenue.

F.7 “Estimates are not watts.”

Correct, and already the rule of this study. A curve, a ledger watt-hour, a CR2032 year count, and a Landauer term on a receipt are estimates or models until a labeled meter returns a reading. Package `measured_j` stays empty. No FPGA board result claimed. What is not an estimate: OpenADR 3.1 and DSX Flex MQTT interoperability (real protocol), Cortex-M4 instruction counts (emulator-measured), SmellNet sequential accuracy (real dataset protocol), and vendor electrical tables used as schedule inputs (datasheet). The future argument does not need those estimates promoted to watts. It needs the envelope to be the first coordinate, and it needs the labeled rows left in their class. A dismissal that says “therefore uncapped hyperscale is the only serious future” changes the subject. Evidence class: **evidence grammar** (§2.7) and Appendix E.

F.8 “Hyperscale water and power are inevitable.”

They are inevitable only for a design whose first coordinate is uncapped import. Metabolic Intelligence refuses that design. Edge intelligence under a metabolic schedule does not contain a cooling plant. A resource-optimized central stays inside a stated megawatt cap and follows the grid, with ordered levers (pause, then speed, then route, then effort, then battery, then shed). Lemma D: if idle import already exceeds the cap, levers that leave nodes powered cannot meet a deep cut. Node sleep is part of feasibility, not a slogan. The civic sentence follows from the envelope, not from a water meter: people should not have to fight over energy to eat or to use AI. Dedicated water, dedicated power, and hyperscale cooling are not required by design at these budgets. Say that. Do not convert it into liters saved. Evidence class: **design consequence** of Propositions 1 and 5 plus the tag schedule model. Not **M** on a field water meter.

Software reference path: no FPGA board result claimed. Estimates are not `measured_j`. Chicken-spoilage Anwar numbers stay internal (no licence).